Sri Lanka Institute of Information Technology

IT3021 – Data Warehousing and Business Intelligence

Assignment 02

MADUGALLE K J W R E W N M R O D

IT23555594

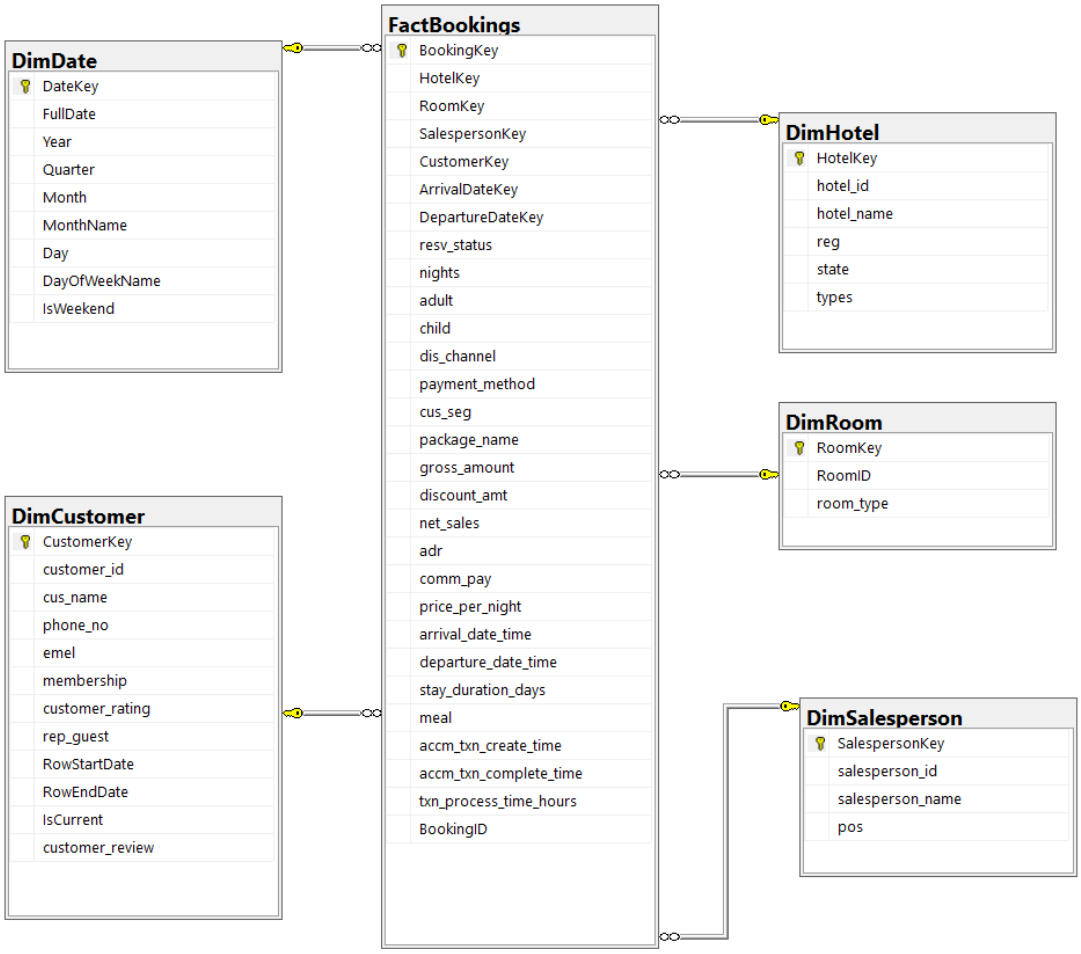
Kandy Uni

Step 1: Data Source for Assignment 2

Description:

The data source for this project is the Hotel_DW Data Warehouse developed in Assignment 01. It is based on a 2024 Hotel Sales dataset containing 6,050 records. The warehouse follows a **Star Schema** architecture to optimise analytical querying and cube processing.

- **Fact Table:** FactBookings (contains quantitative metrics like Net Sales, ADR, and Stay Duration).
- **Dimensions:** DimHotel, DimDate, DimCustomer, DimRoom, and DimSalesperson.

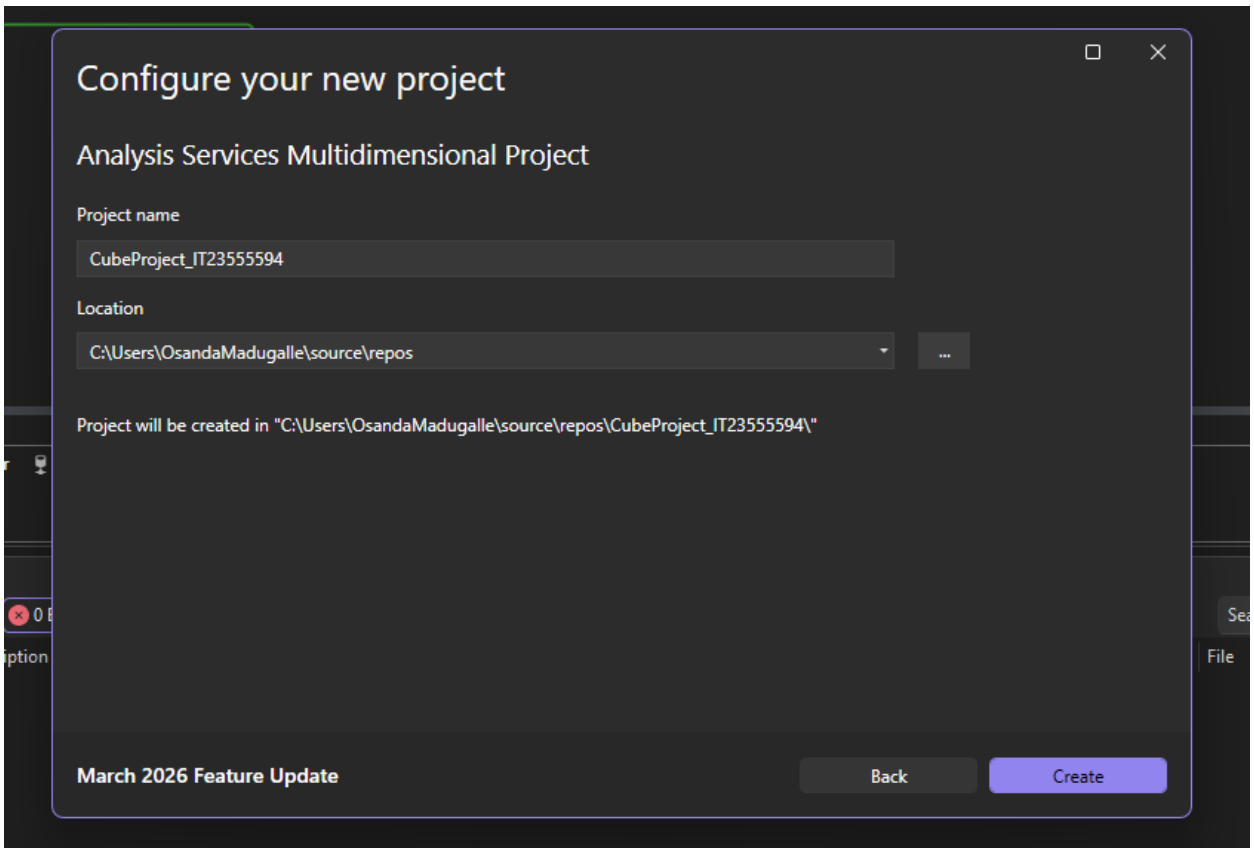


Step 2: SSAS Cube Implementation

2.1. Creating the Analysis Services Project

Description:

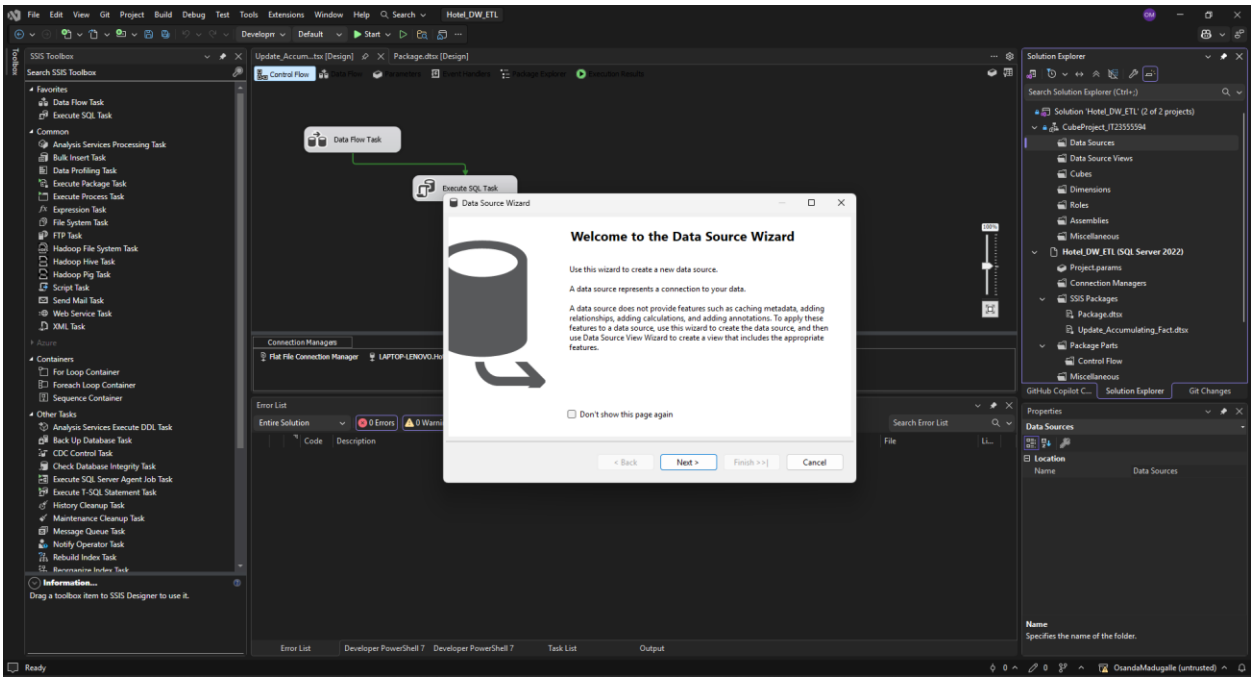
The development began by launching **Visual Studio** and creating a new project using the **Analysis Services Multidimensional and Data Mining** template. This project was named CubeProject_IT23555594.

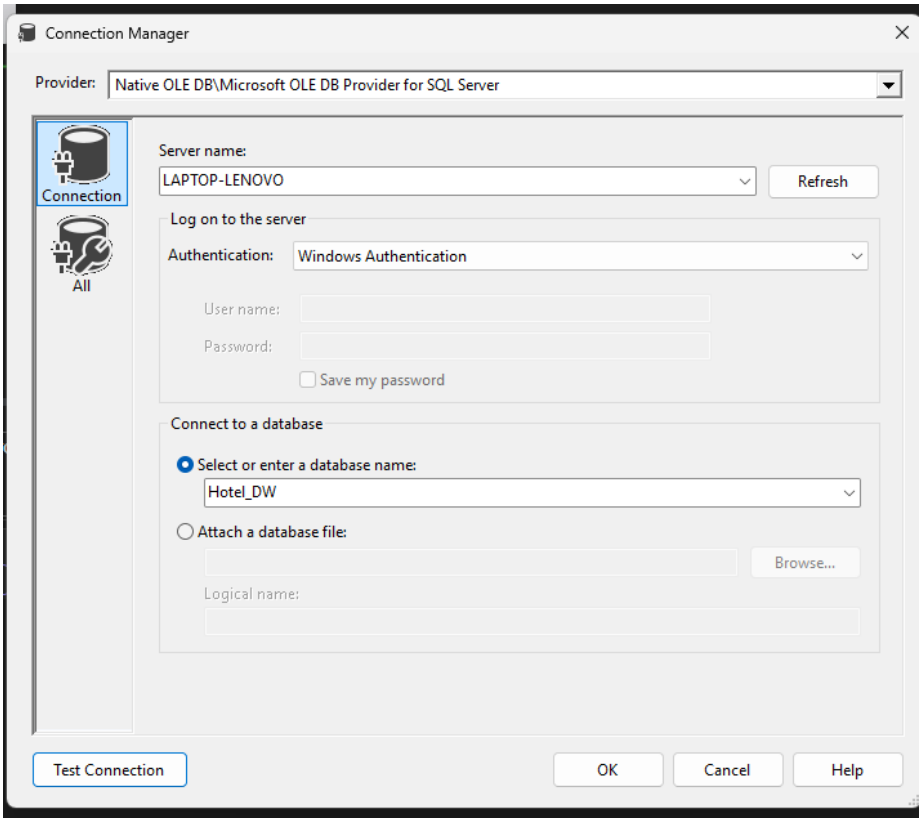


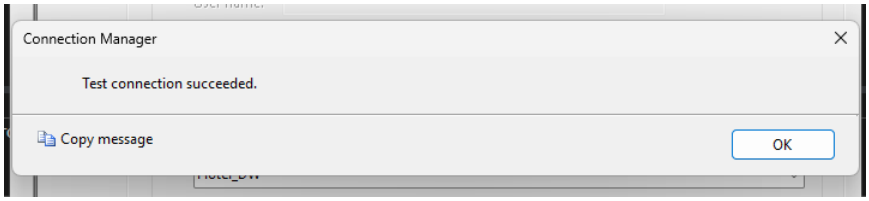
2.2. Defining the Data Source

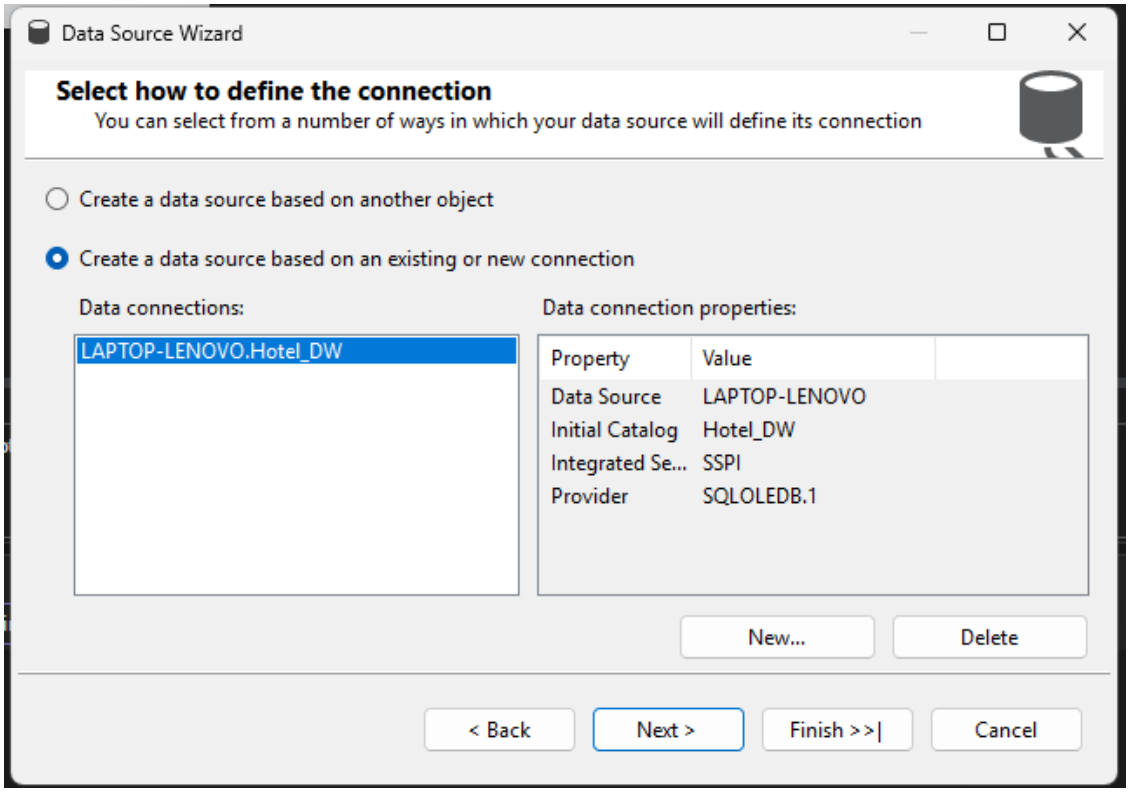
Description:

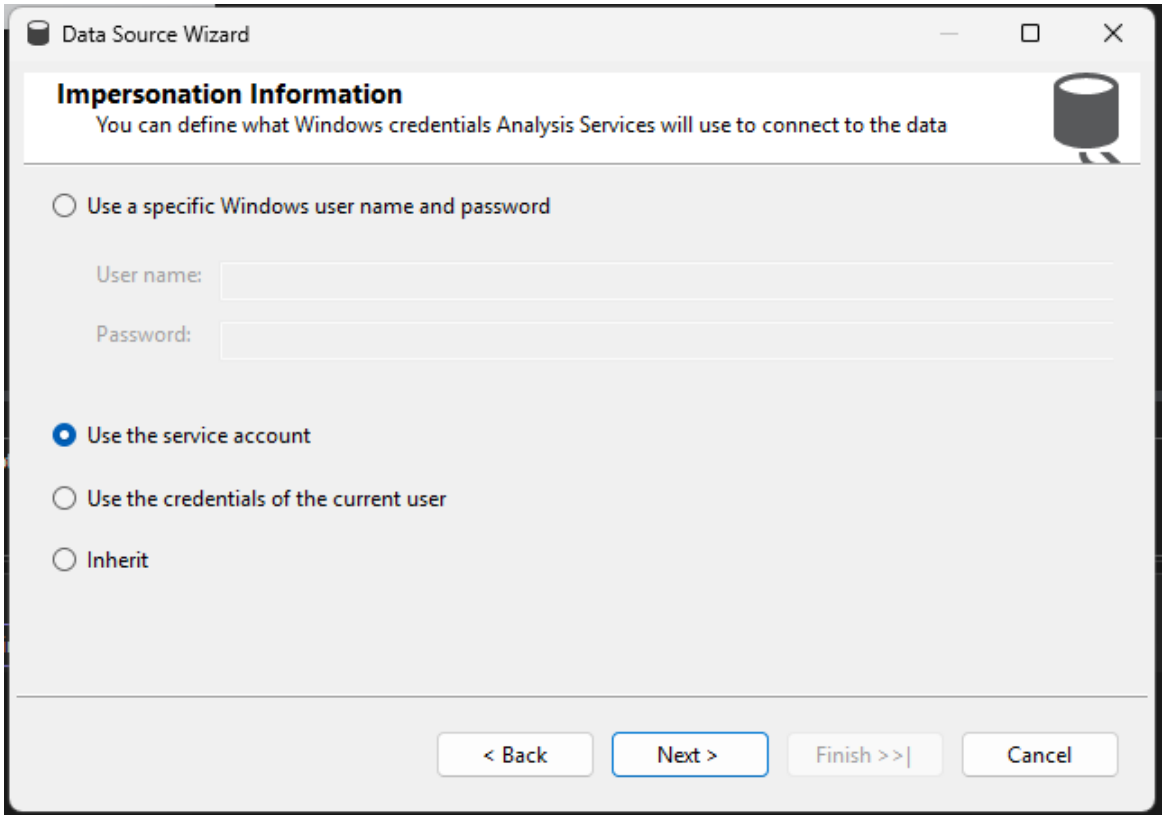
A connection to the Hotel_DW database was established. The provider used was **Native OLE DB\Microsoft OLE DB Provider for SQL Server**. The server name was set to LAPTOP-LENOVO, and the impersonation information was set to use the Service Account to ensure the cube has persistent access to the data warehouse.

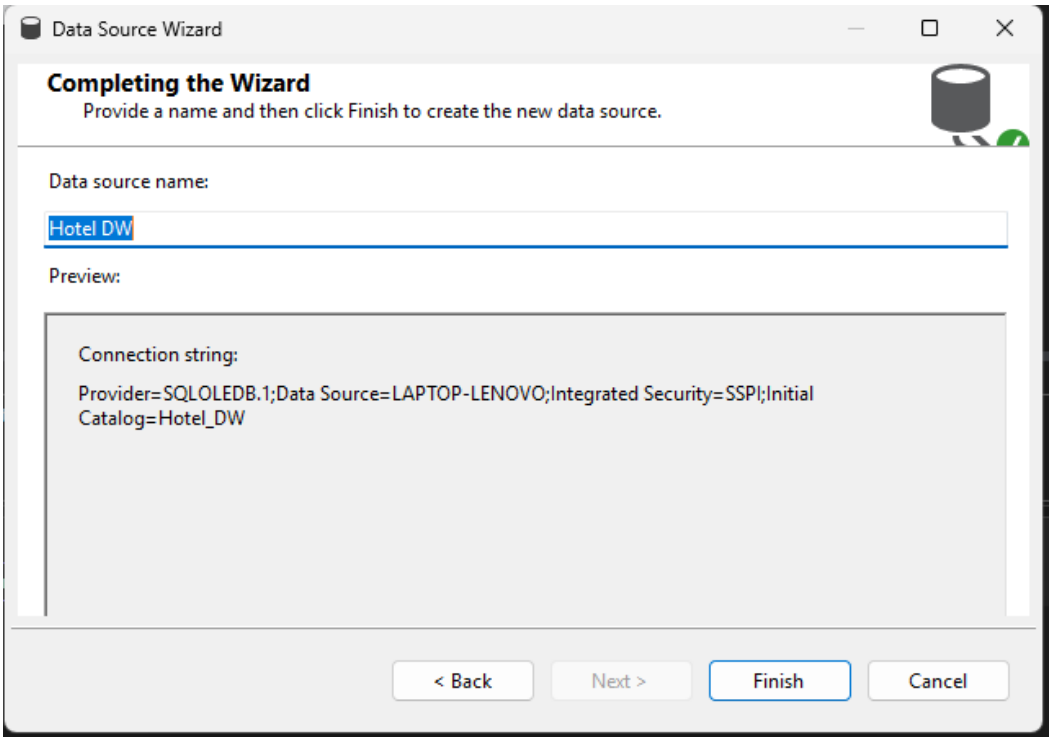

Data Source Wizard
Welcome to the Data Source Wizard
Use this wizard to create a new data source.
A data source represents a connection to your data.
A data source does not provide features such as caching metadata, adding relationships, adding calculations, and adding annotations. To apply these features to a data source, use this wizard to create the data source, and then use Data Source View Wizard to create a view that includes the appropriate features.
Don't show this page again
< Back
Next >
Finish >>|
Cancel


Connection Manager
Provider:
Native OLE DB\Microsoft OLE DB Provider for SQL Server
Connection
All
Server name:
LAPTOP-LENOVO
Refresh
Log on to the server
Authentication:
Windows Authentication
User name:
Password:
Save my password
Connect to a database
Select or enter a database name:
Hotel_DW
Attach a database file:
Browse...
Logical name:
Test Connection
OK
Cancel
Help


Connection Manager
Test connection succeeded.
Copy message
OK


Data Source Wizard
Select how to define the connection
You can select from a number of ways in which your data source will define its connection
Create a data source based on another object
Create a data source based on an existing or new connection
Data connections:
LAPTOP-LENOVO.Hotel_DW
Data connection properties:
Property
Value
Data Source
LAPTOP-LENOVO
Initial Catalog
Hotel_DW
Integrated Se...
SSPI
Provider
SQLOLEDB.1
New...
Delete
< Back
Next >
Finish >>|
Cancel

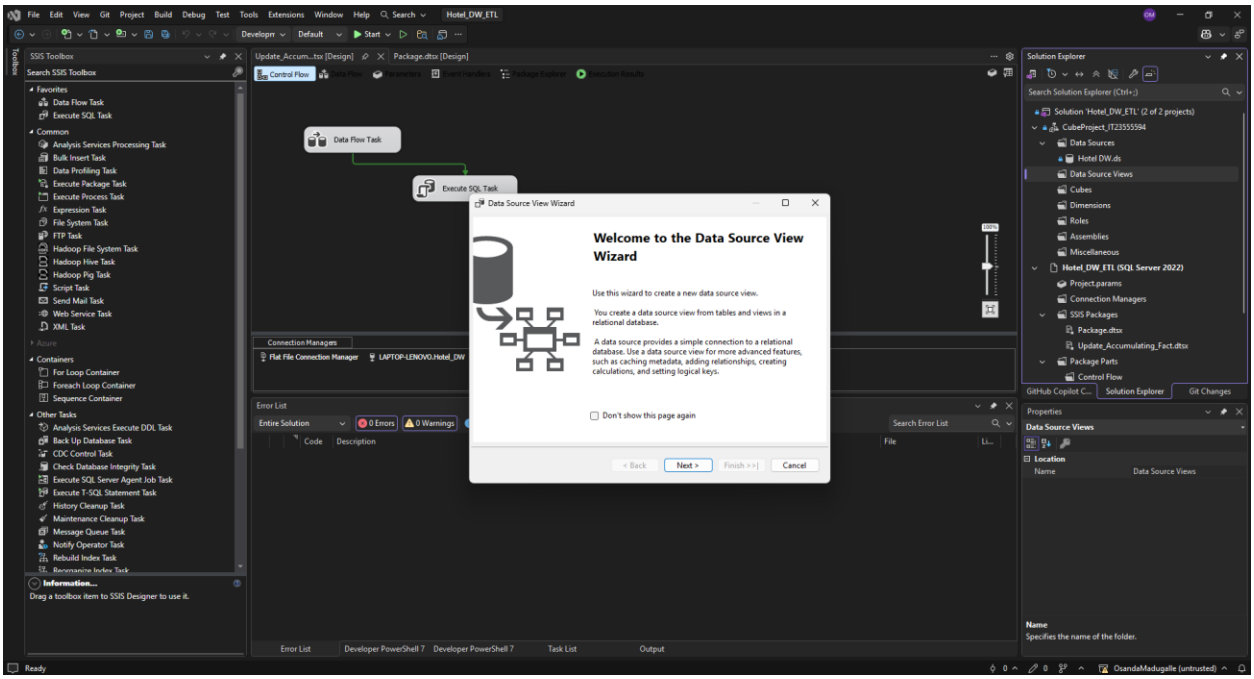

Data Source Wizard
Impersonation Information
You can define what Windows credentials Analysis Services will use to connect to the data
Use a specific Windows user name and password
User name:
Password:
Use the service account
Use the credentials of the current user
Inherit
< Back
Next >
Finish >>|
Cancel

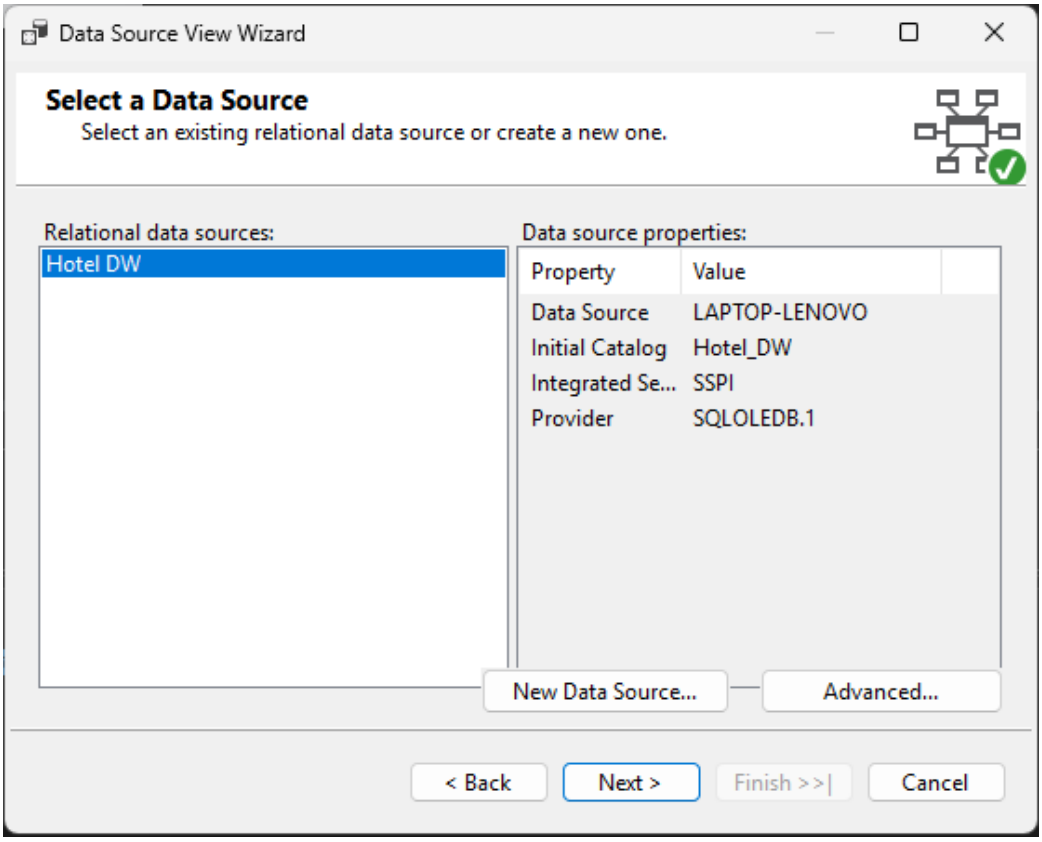


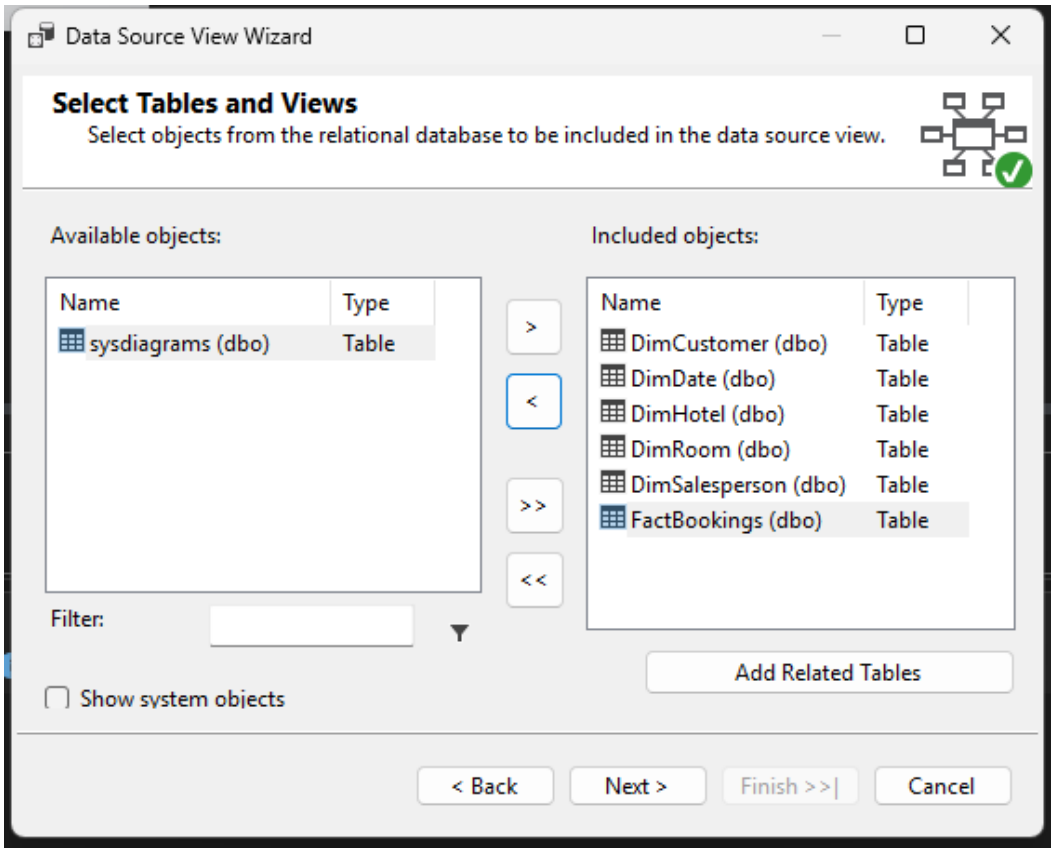
2.3. Creating the Data Source View (DSV)

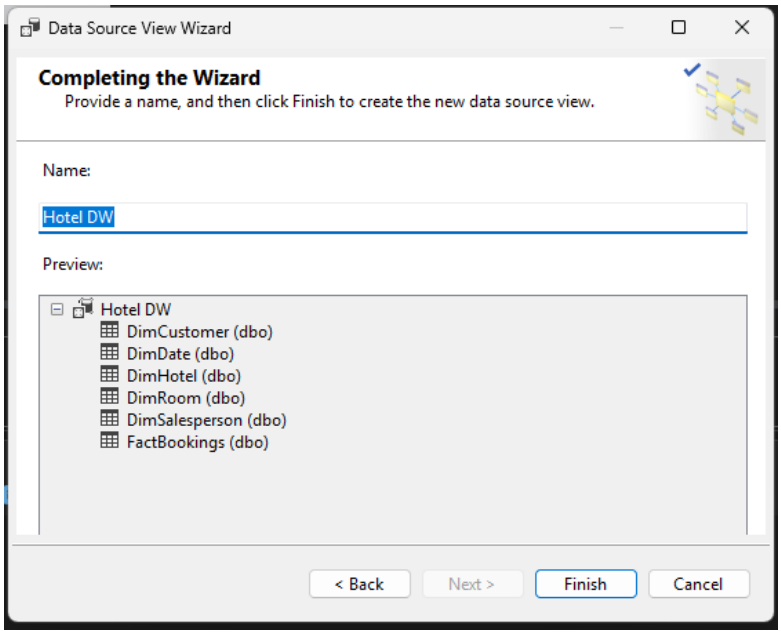
Description:

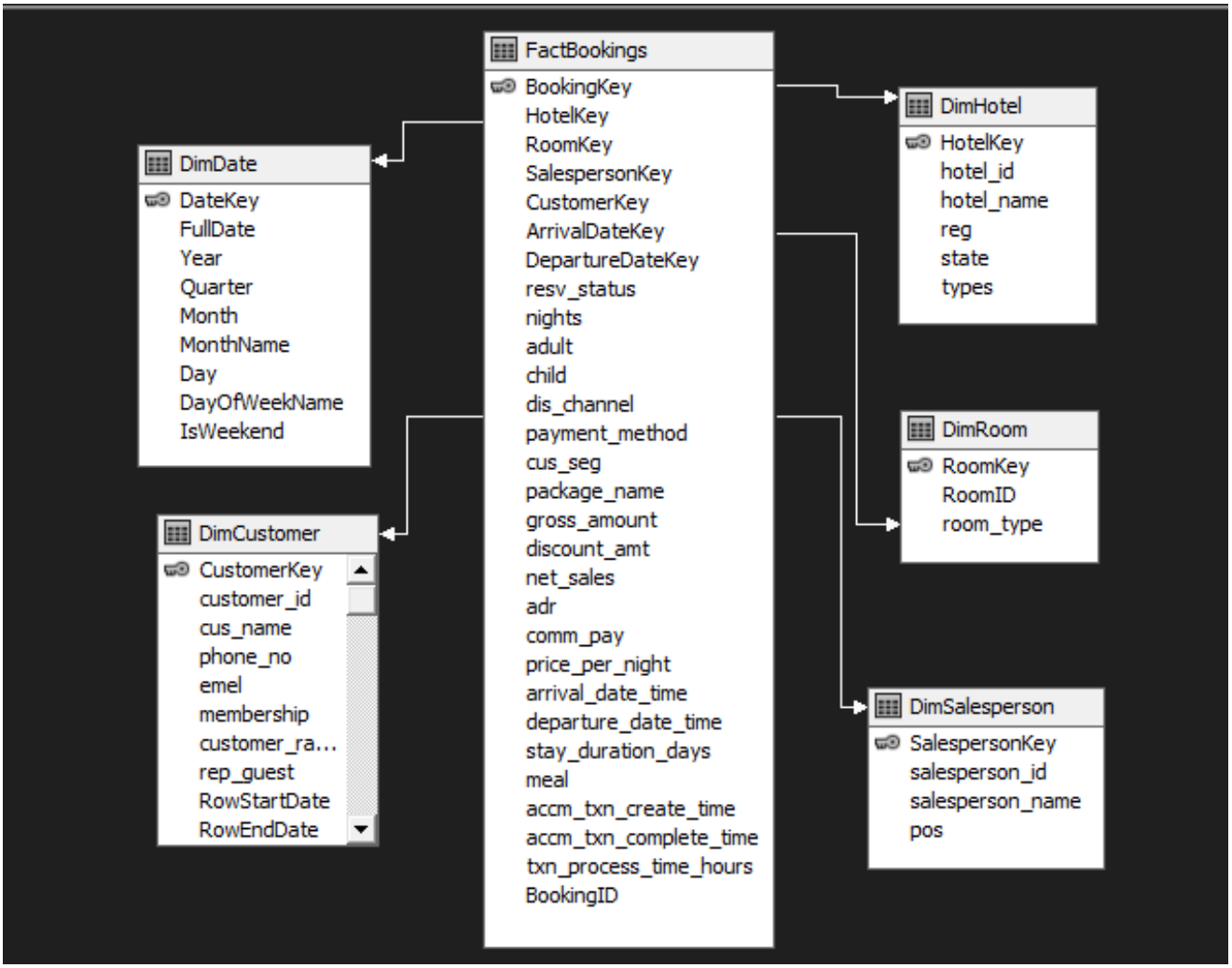
The Data Source View was created to define the schema for the cube. All tables from the Hotel_DW (FactBookings and the 5 Dimensions) were added to the DSV. This layer allows for the definition of logical primary keys and relationships without altering the physical database.




Data Source View Wizard
Select a Data Source
Select an existing relational data source or create a new one.
Relational data sources:
Hotel DW
Data source properties:
Property
Value
Data Source
LAPTOP-LENOVO
Initial Catalog
Hotel_DW
Integrated Se...
SSPI
Provider
SQLOLEDB.1
New Data Source...
Advanced...
< Back
Next >
Finish >>|
Cancel


Data Source View Wizard
Select Tables and Views
Select objects from the relational database to be included in the data source view.
Available objects:
Name
Type
sysdiagrams (dbo)
Table
Included objects:
Name
Type
DimCustomer (dbo)
Table
DimDate (dbo)
Table
DimHotel (dbo)
Table
DimRoom (dbo)
Table
DimSalesperson (dbo)
Table
FactBookings (dbo)
Table
>
<
>>
<<
Filter:
Show system objects
Add Related Tables
< Back
Next >
Finish >>|
Cancel

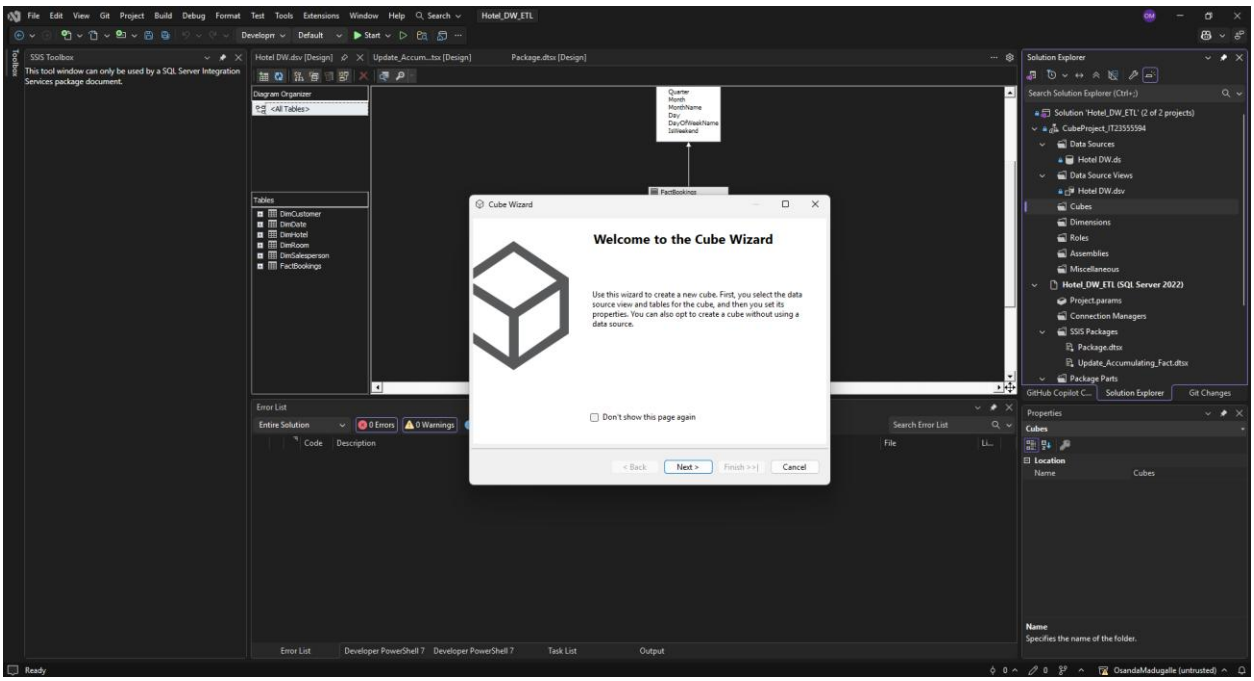

Data Source View Wizard
Completing the Wizard
Provide a name, and then click Finish to create the new data source view.
Name:
Hotel DW
Preview:
Hotel DW
DimCustomer (dbo)
DimDate (dbo)
DimHotel (dbo)
DimRoom (dbo)
DimSalesperson (dbo)
FactBookings (dbo)
< Back
Next >
Finish
Cancel

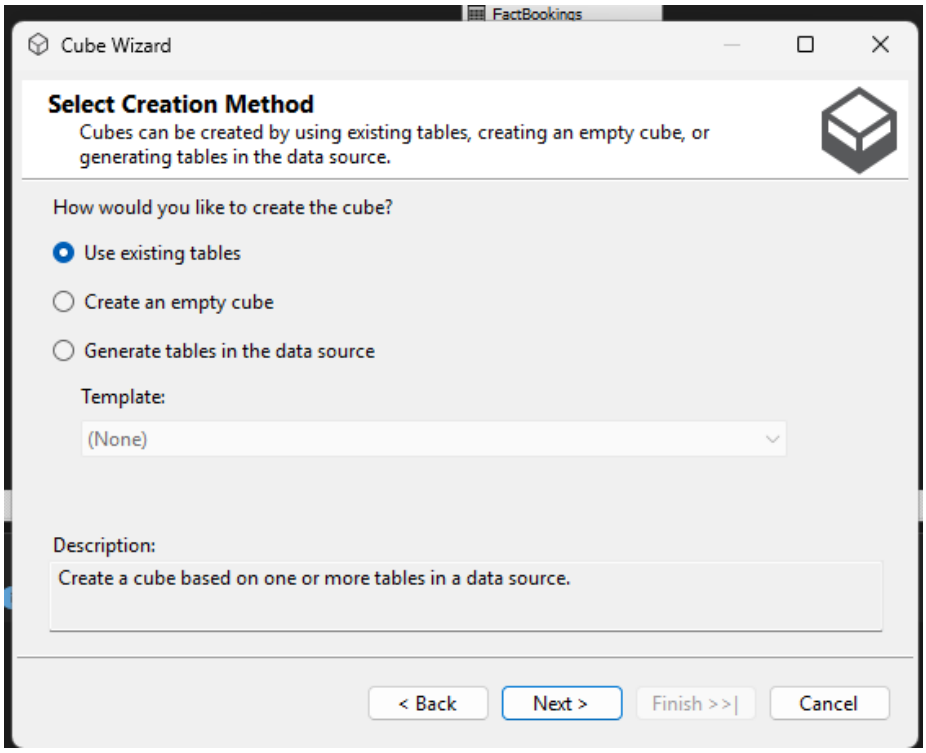

FactBookings
BookingKey
HotelKey
RoomKey
SalespersonKey
CustomerKey
ArrivalDateKey
DepartureDateKey
resv_status
nights
adult
child
dis_channel
payment_method
cus_seg
package_name
gross_amount
discount_amt
net_sales
adr
comm_pay
price_per_night
arrival_date_time
departure_date_time
stay_duration_days
meal
accm_txn_create_time
accm_txn_complete_time
txn_process_time_hours
BookingID
DimDate
DateKey
FullDate
Year
Quarter
Month
MonthName
Day
DayOfWeekName
IsWeekend
DimCustomer
CustomerKey
customer_id
cus_name
phone_no
emel
membership
customer_ra...
rep_guest
RowStartDate
RowEndDate
DimHotel
HotelKey
hotel_id
hotel_name
reg
state
types
DimRoom
RoomKey
RoomID
room_type
DimSalesperson
SalespersonKey
salesperson_id
salesperson_name
pos

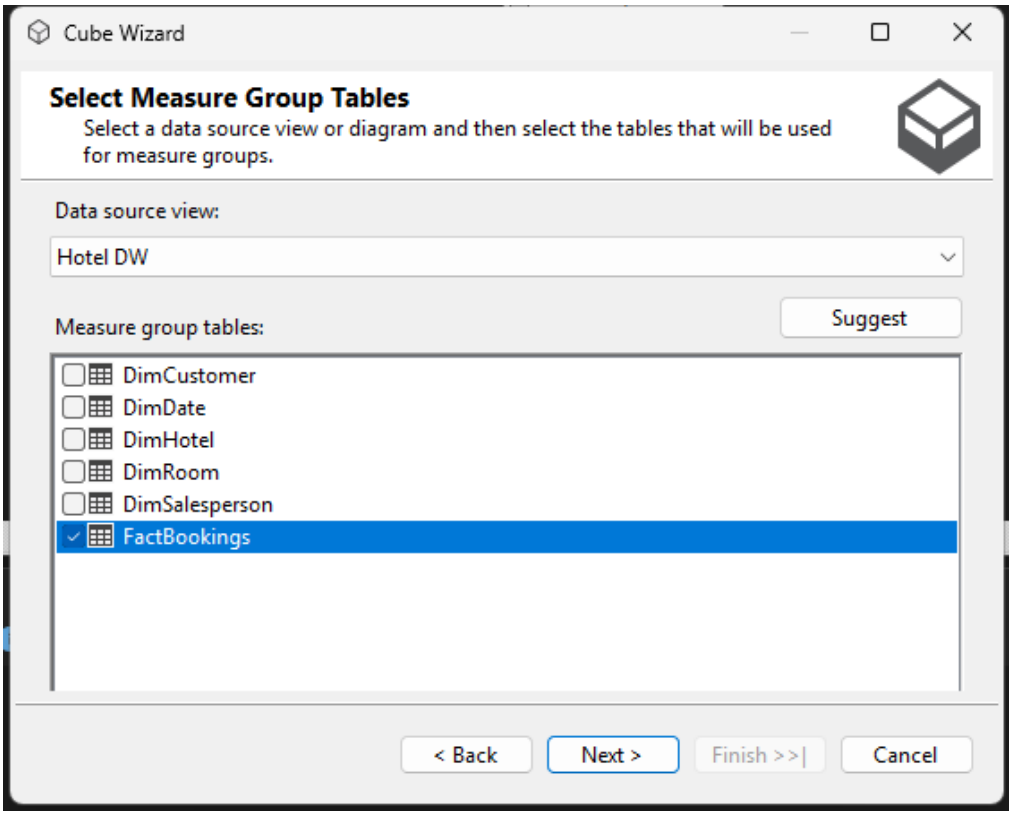
2.4. Creating the Cube and Measure Groups

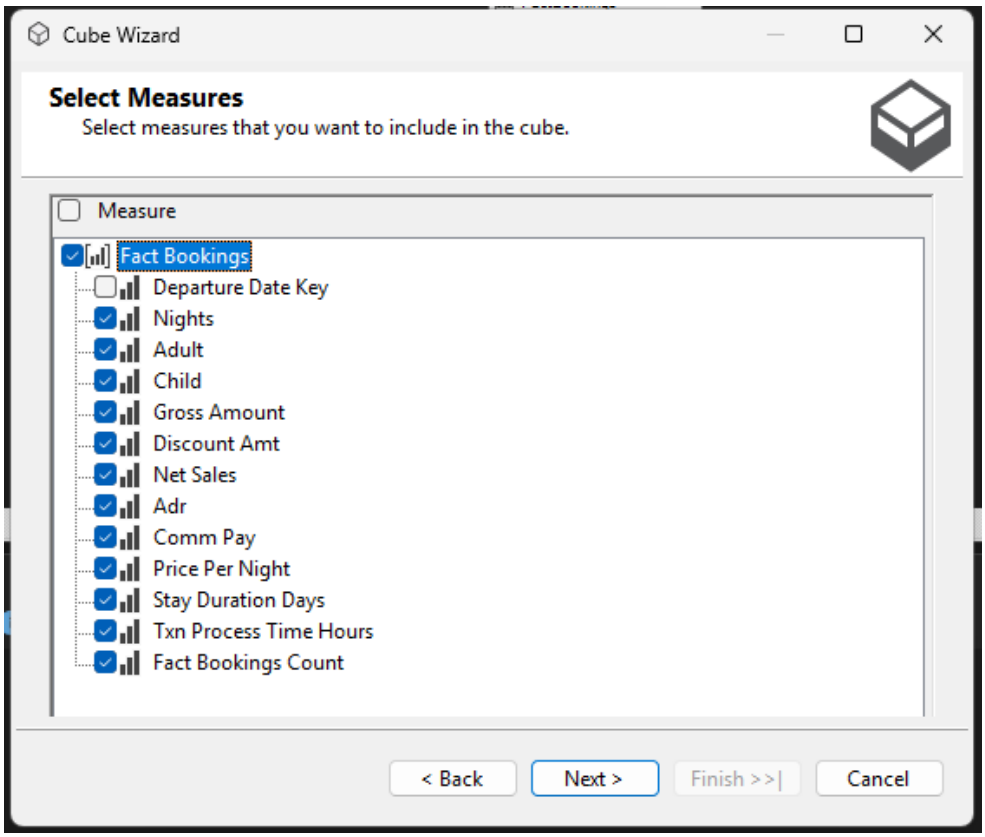
Description:

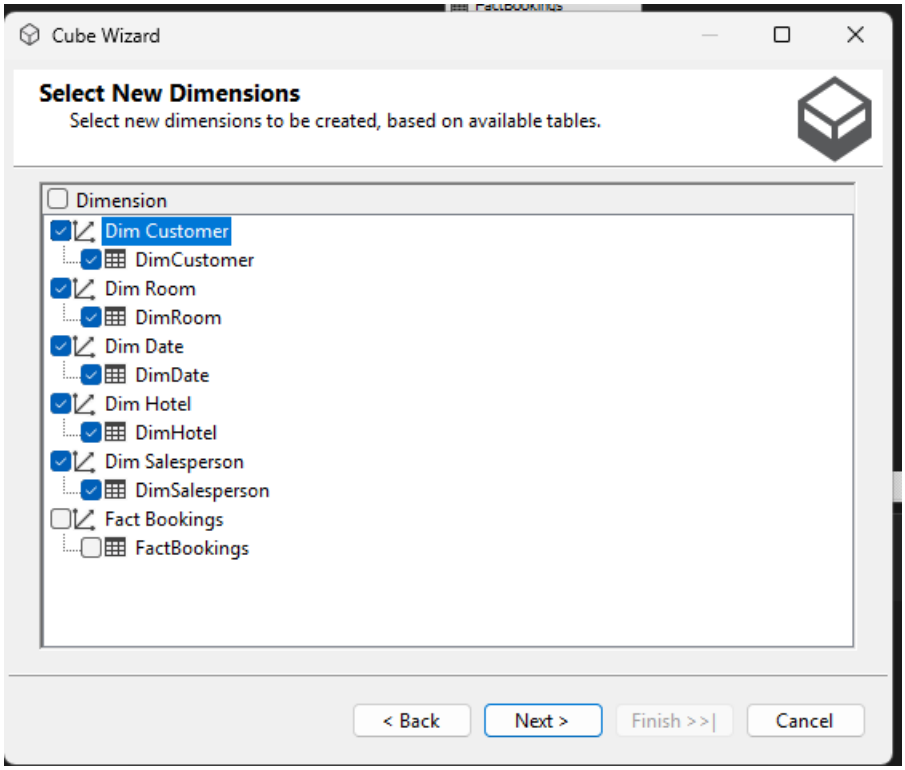
The Cube Wizard was used to initialise the OLAP structure. FactBookings was identified as the primary **Measure Group**. During this stage, the base measures were defined, including Net Sales, Gross Amount, and Booking Count. The wizard automatically detected the relationships between the Fact table and the Dimensions based on the Primary/Foreign key constraints defined in the Hotel_DW.

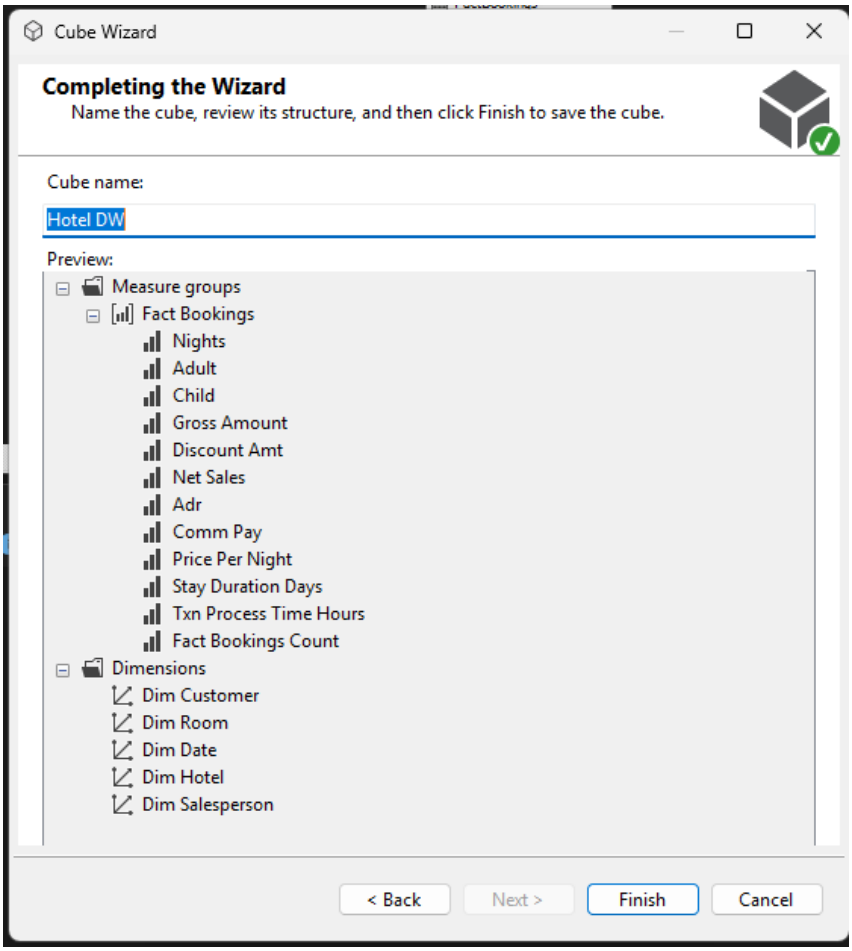


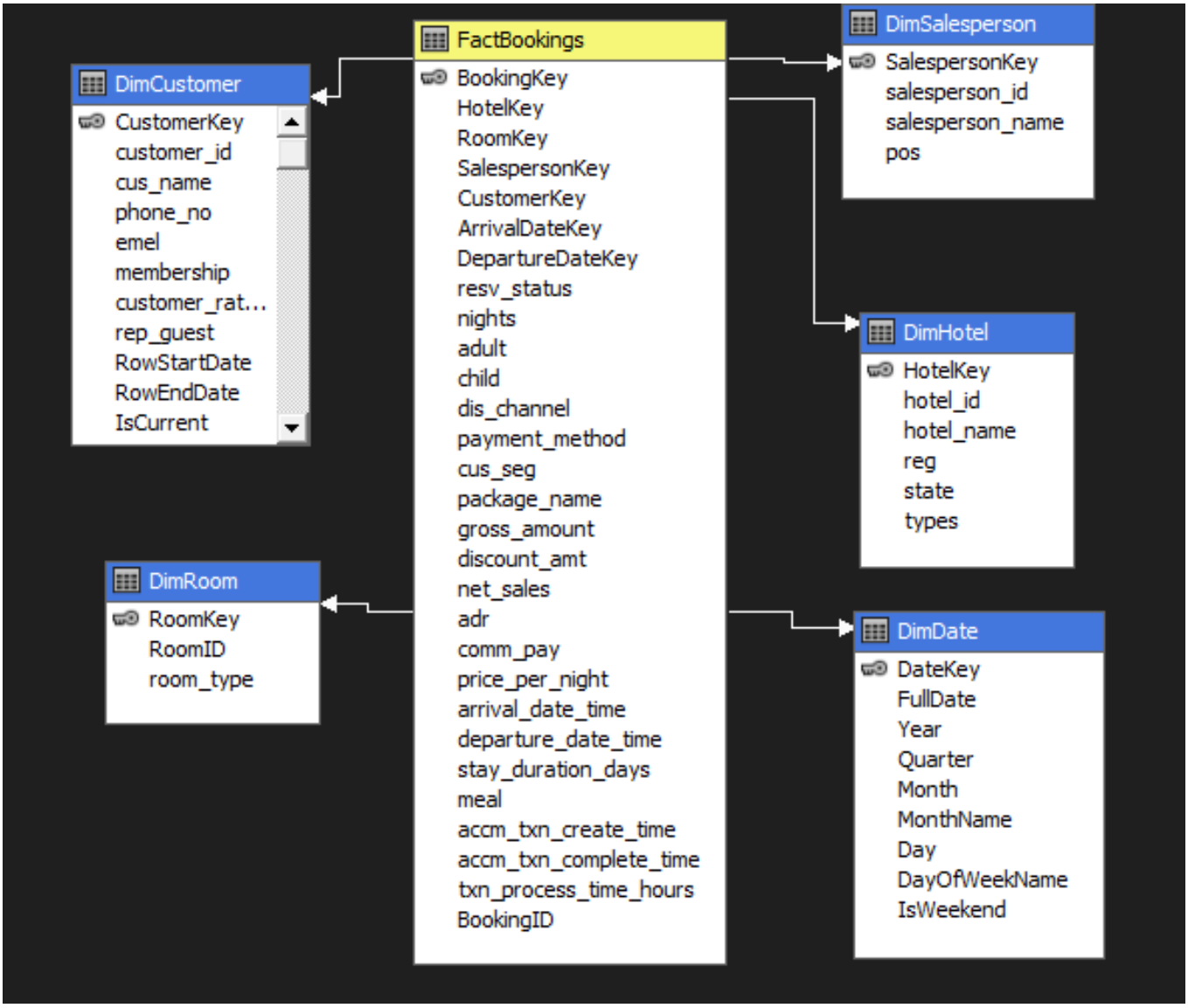



Cube Wizard
Select Measure Group Tables
Select a data source view or diagram and then select the tables that will be used for measure groups.
Data source view:
Hotel DW
Measure group tables:
Suggest
DimCustomer
DimDate
DimHotel
DimRoom
DimSalesperson
FactBookings
< Back
Next >
Finish >>|
Cancel


Cube Wizard
Select Measures
Select measures that you want to include in the cube.
Measure
Fact Bookings
Departure Date Key
Nights
Adult
Child
Gross Amount
Discount Amt
Net Sales
Adr
Comm Pay
Price Per Night
Stay Duration Days
Txn Process Time Hours
Fact Bookings Count
< Back
Next >
Finish >>|
Cancel


Cube Wizard
Select New Dimensions
Select new dimensions to be created, based on available tables.
Dimension
Dim Customer
DimCustomer
Dim Room
DimRoom
Dim Date
DimDate
Dim Hotel
DimHotel
Dim Salesperson
DimSalesperson
Fact Bookings
FactBookings
< Back
Next >
Finish >>|
Cancel


Cube Wizard
Completing the Wizard
Name the cube, review its structure, and then click Finish to save the cube.
Cube name:
Hotel DW
Preview:
Measure groups
Fact Bookings
Nights
Adult
Child
Gross Amount
Discount Amt
Net Sales
Adr
Comm Pay
Price Per Night
Stay Duration Days
Txn Process Time Hours
Fact Bookings Count
Dimensions
Dim Customer
Dim Room
Dim Date
Dim Hotel
Dim Salesperson
< Back
Next >
Finish
Cancel

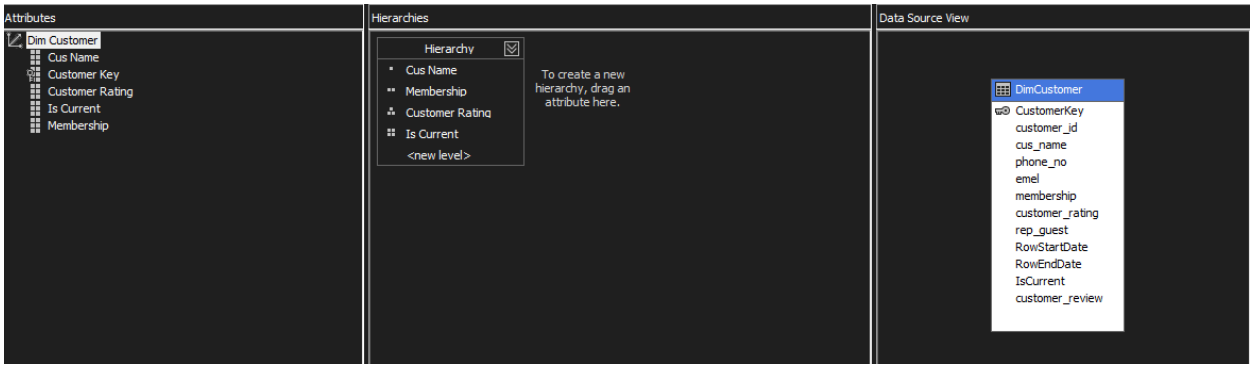


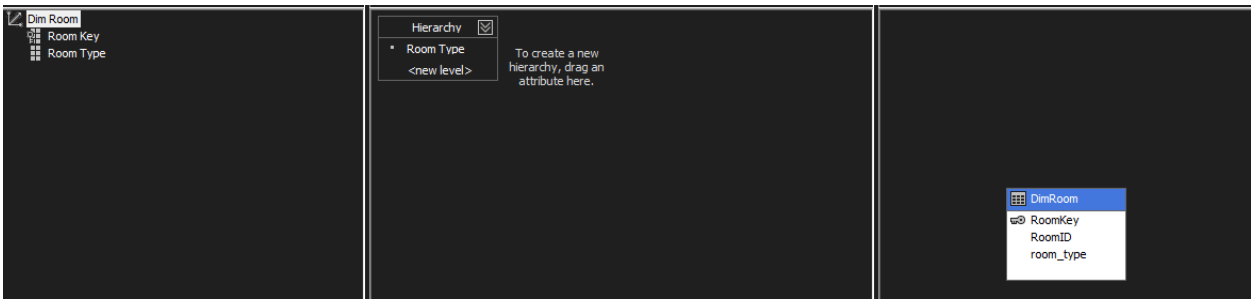
2.5. Enhancing Dimensions and Building Hierarchies

Description:

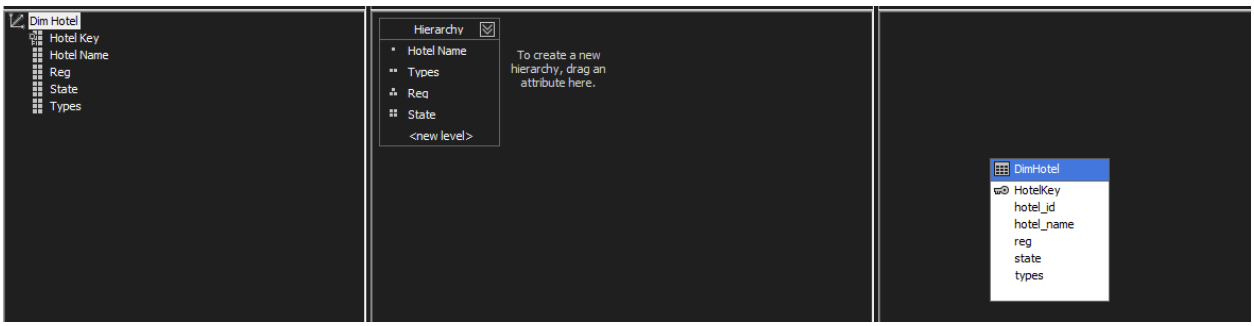
After the initial cube was created, each dimension was individually refined in the **Dimension Designer**. To support advanced analytical requirements such as temporal trends, a custom **Time Hierarchy** was built within the DimDate dimension.

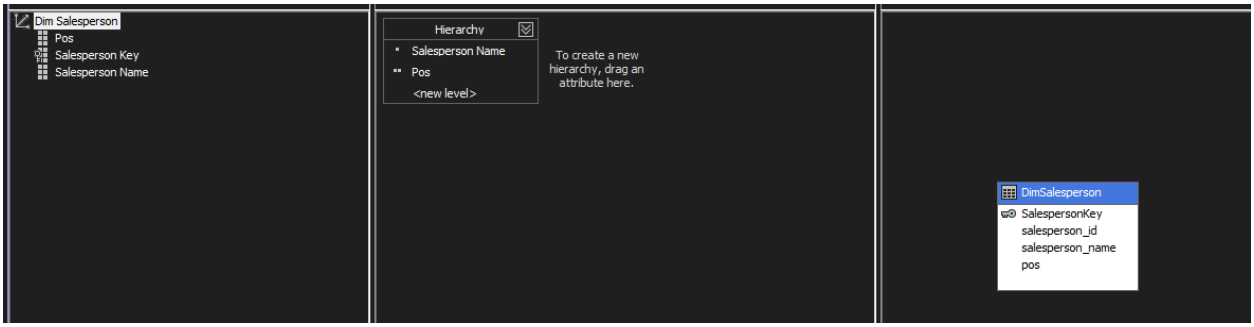
- **Attributes Added:** Year, Quarter, Month, and Day.
- **Hierarchy Structure:** Year $\rightarrow$ Quarter $\rightarrow$ Month $\rightarrow$ Day.
- **Benefit:** This structure allows the Power BI reports to perform seamless drill-down operations from annual overviews to daily transaction details.


Attributes
Dim Customer
Cus Name
Customer Key
Customer Rating
Is Current
Membership
Hierarchies
Hierarchy
Cus Name
Membership
Customer Rating
Is Current
<new level>
To create a new hierarchy, drag an attribute here.
Data Source View
DimCustomer
CustomerKey
customer_id
cus_name
phone_no
emel
membership
customer_rating
rep_guest
RowStartDate
RowEndDate
IsCurrent
customer_review


Dim Room
Room Key
Room Type
Hierarchy
Room Type
<new level>
To create a new hierarchy, drag an attribute here.
DimRoom
RoomKey
RoomID
room_type

Dim Date
Date Key
Day
Month
Quarter
Year
Hierarchy
Year
Quarter
Month
Day
<new level>
To create a new hierarchy, drag an attribute here.
DimDate
DateKey
FullDate
Year
Quarter
Month
MonthName
Day
DayOfWeekName
IsWeekend

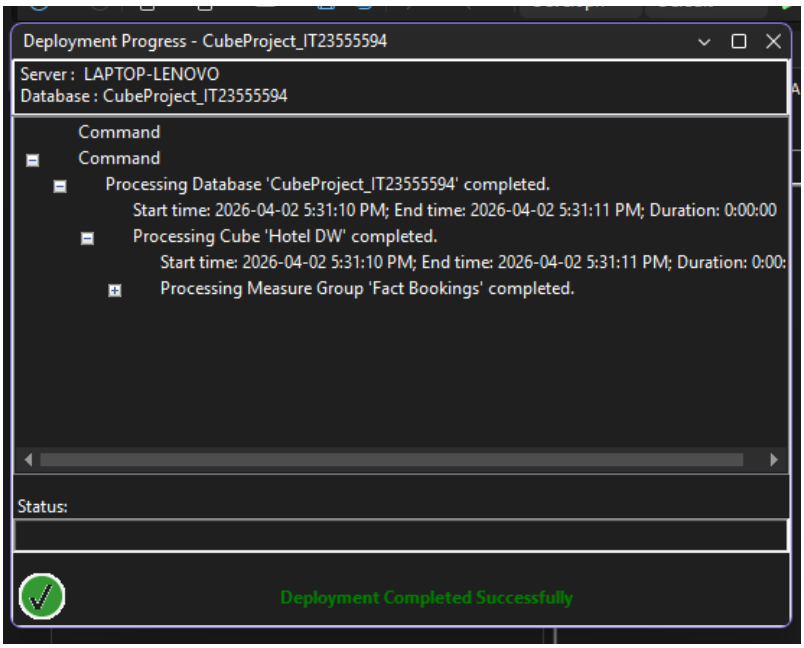

Dim Hotel
Hotel Key
Hotel Name
Reg
State
Types
Hierarchy
Hotel Name
Types
Reg
State
<new level>
To create a new hierarchy, drag an attribute here.
DimHotel
HotelKey
hotel_id
hotel_name
reg
state
types


Dim Salesperson
Pos
Salesperson Key
Salesperson Name
Hierarchy
Salesperson Name
Pos
<new level>
To create a new hierarchy, drag an attribute here.
DimSalesperson
SalespersonKey
salesperson_id
salesperson_name
pos

2.6. Deployment and Processing

Description:

The final step involved deploying the solution to the **Analysis Services Server instance**. This compiles the XMLA scripts and sends them to the server. Once deployed, the cube was "Processed" to fetch the 6,050 records from the Hotel_DW and build the multi-dimensional storage.



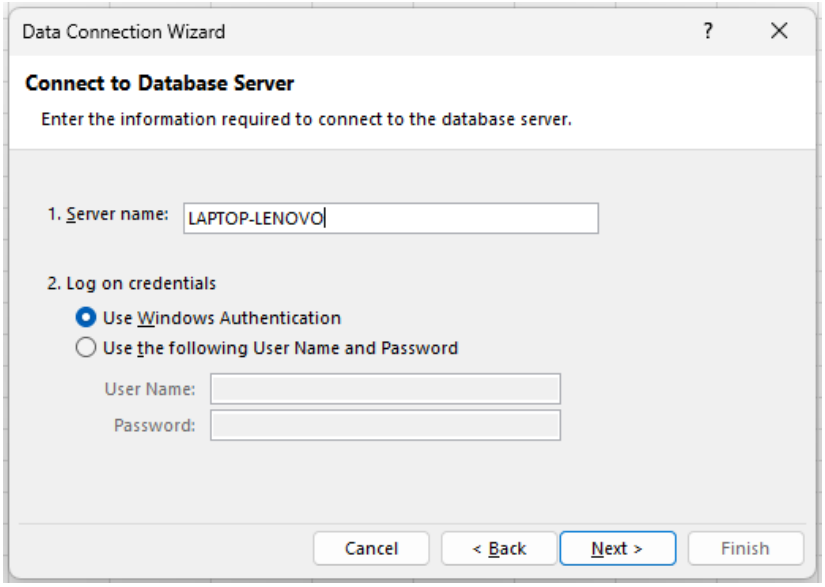
Step 3: Demonstration of OLAP Operations (Excel)

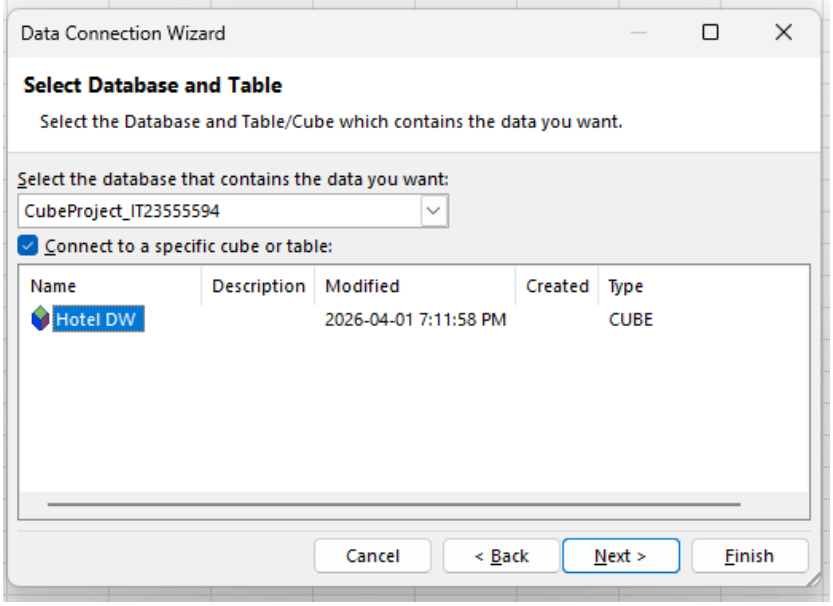
Description:

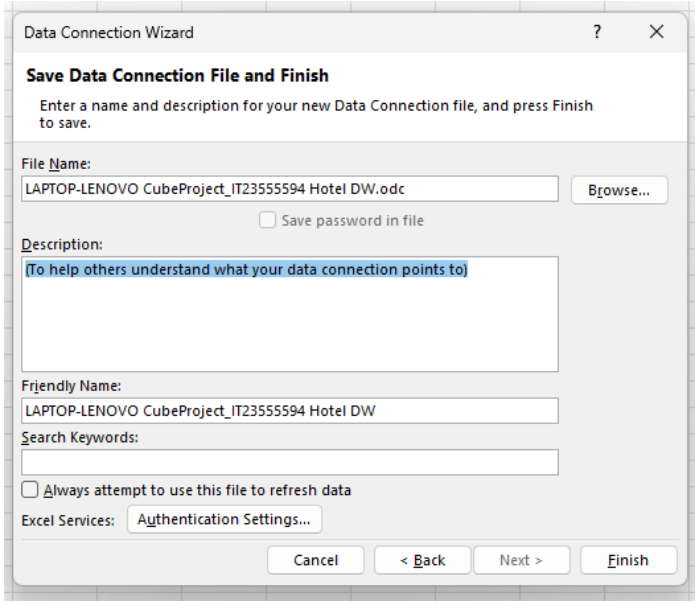
To validate the multi-dimensional analytical capabilities of the deployed SSAS Cube, a connection was established using Microsoft Excel as the front-end client. This allows for real-time interaction with the Hotel_DW measures and dimensions via an OLAP pivot connection.

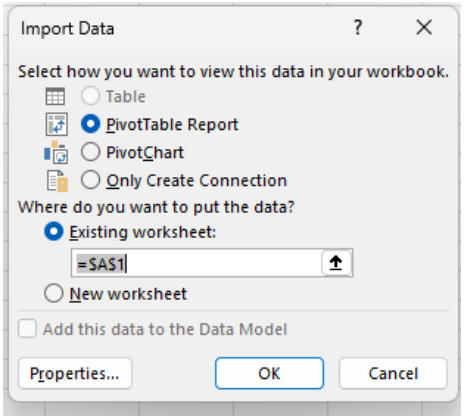
3.1. Connecting to the SSAS Provider

- **Action:** In Excel, the connection was initialised via **Data > Get Data > From Database > From Analysis Services**.
- **Server:** LAPTOP-LENOVO.
- **Database:** CubeProject_IT23555594.







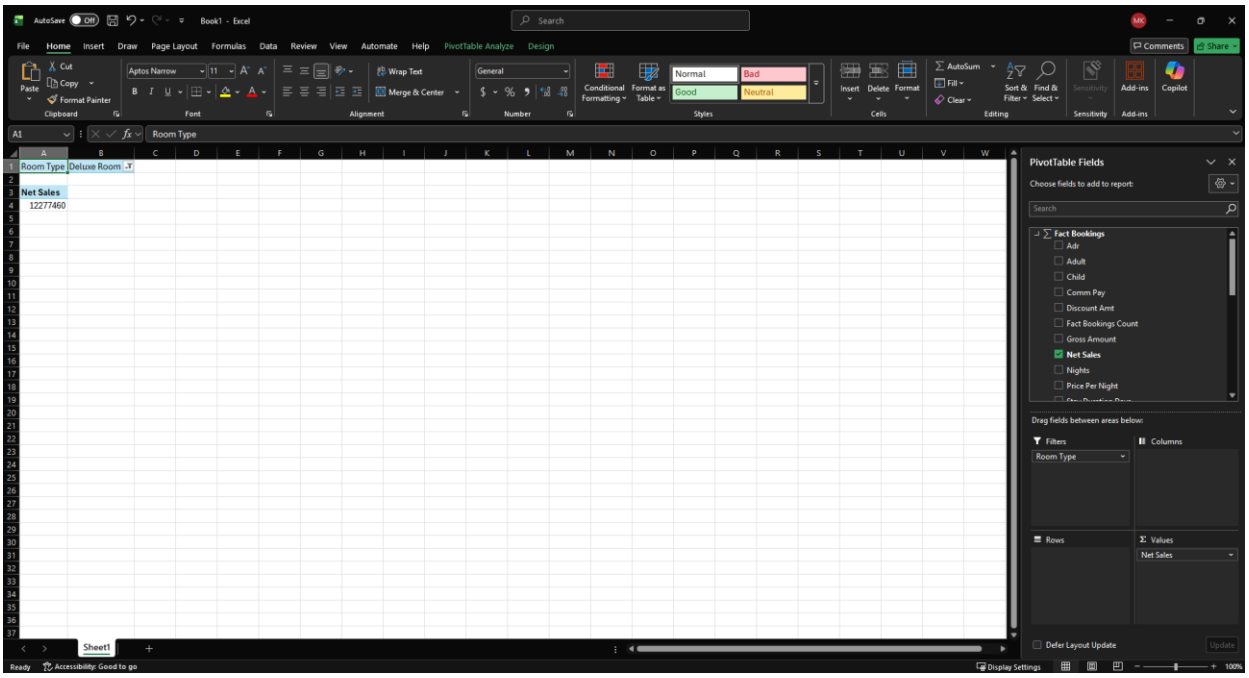


3.2. Executing OLAP Operations

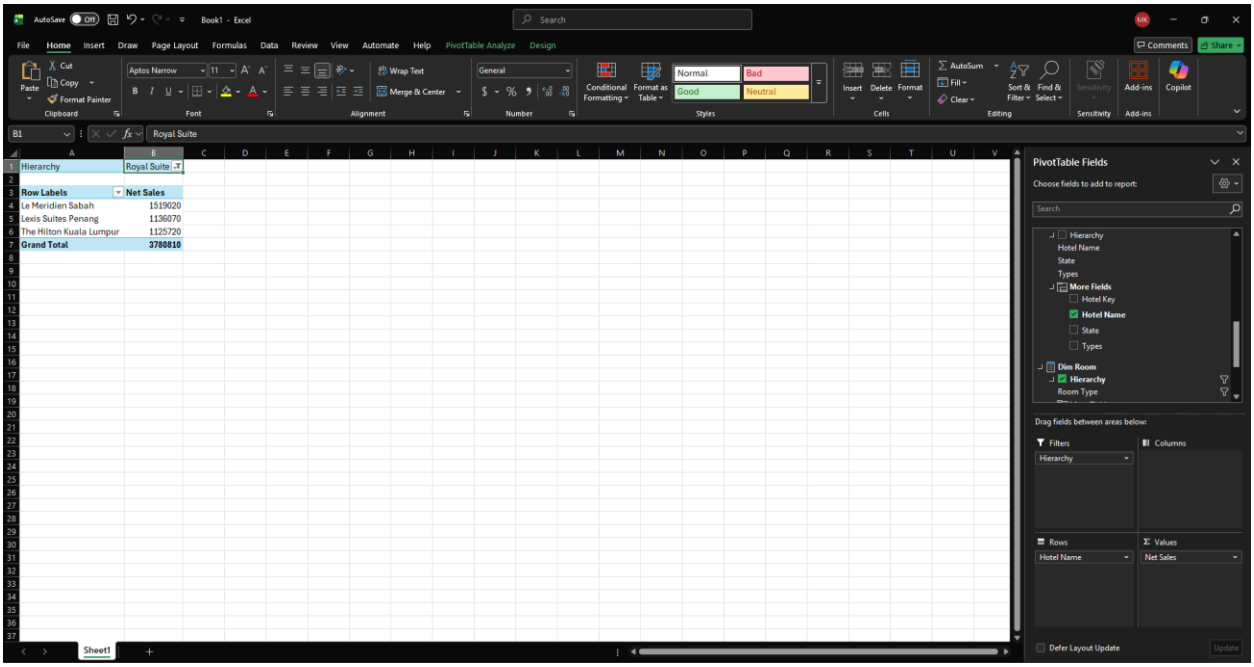
Operation	Technical Action Taken	Business Insight
Slice	Filtered the Room Type dimension to show only "Single" rooms.	Isolates revenue for a specific product category.
Dice	Added a second filter for a specific Hotel Name.	View a specific cross-section (e.g., Single rooms in the branch).
Drill-Down	Expanded the Year attribute in the Time Hierarchy to show Quarters.	Breaks down annual totals into seasonal performance.
Roll-Up	Collapsed the Month level back up to the Quarter level.	Summarises detailed data into a high-level executive view.
Pivot	Swapped Hotel Name from Rows to Columns.	Changes the orientation of the report for better comparison.

3.3. Screenshots of Operations

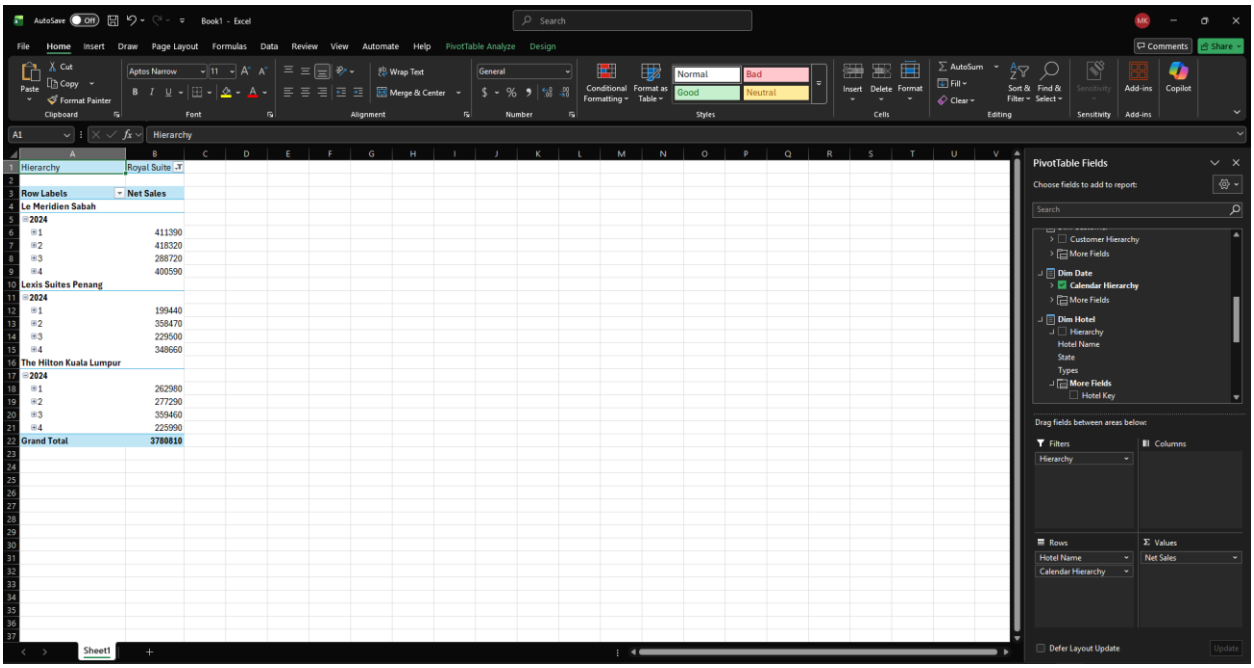
I. Slice Operation



II. Dice Operation

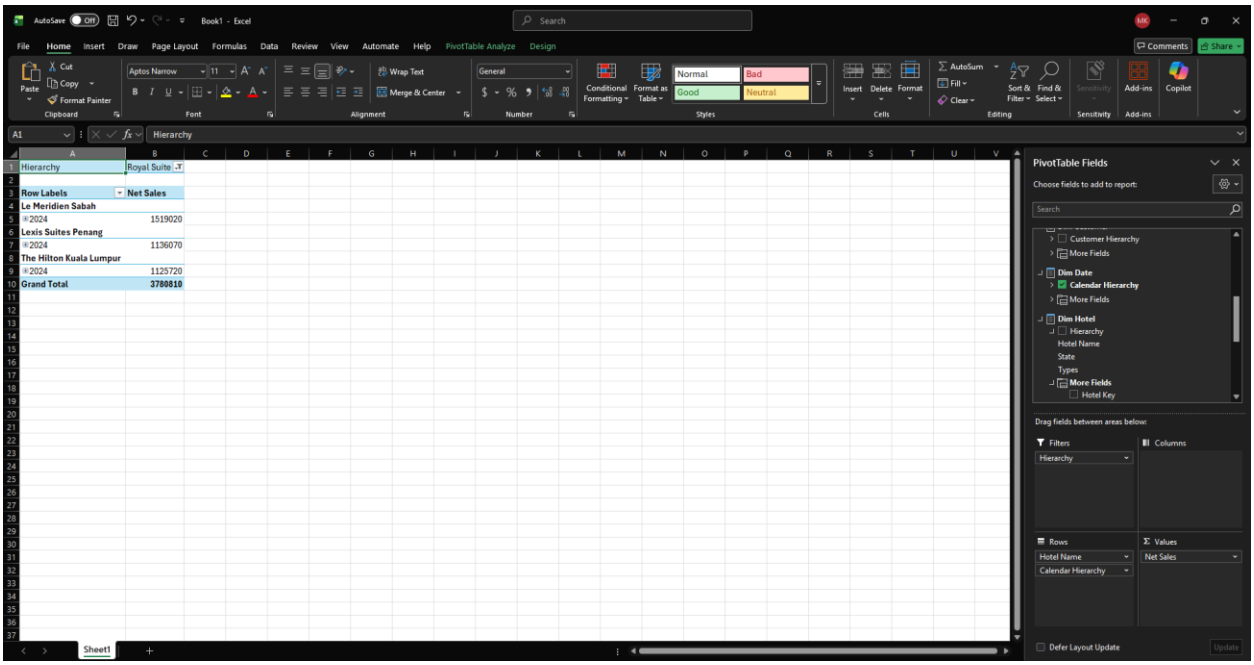


III. Drill-Down Operation



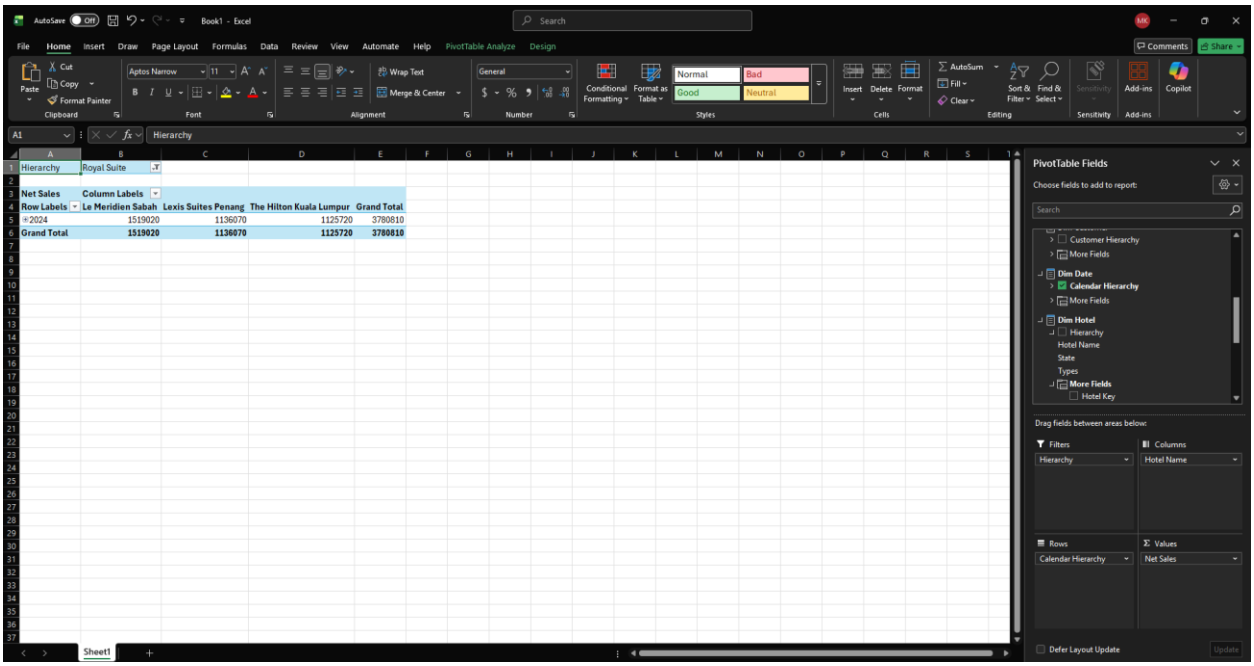
Hierarchy	Royal Suite
Row Labels	Net Sales
Le Meridien Sabah	
2024	
1	411390
2	418320
3	288720
4	400590
Lexis Suites Penang	
2024	
1	199440
2	358470
3	229500
4	348660
The Hilton Kuala Lumpur	
2024	
1	262980
2	277290
3	359460
4	225990
Grand Total	3780810

IV. Roll-Up



Hierarchy	Royal Suite
Row Labels	Net Sales
Le Meridien Sabah	
2024	1519020
Lexis Suites Penang	
2024	1136070
The Hilton Kuala Lumpur	
2024	1125720
Grand Total	3780810

V. Pivot Operation



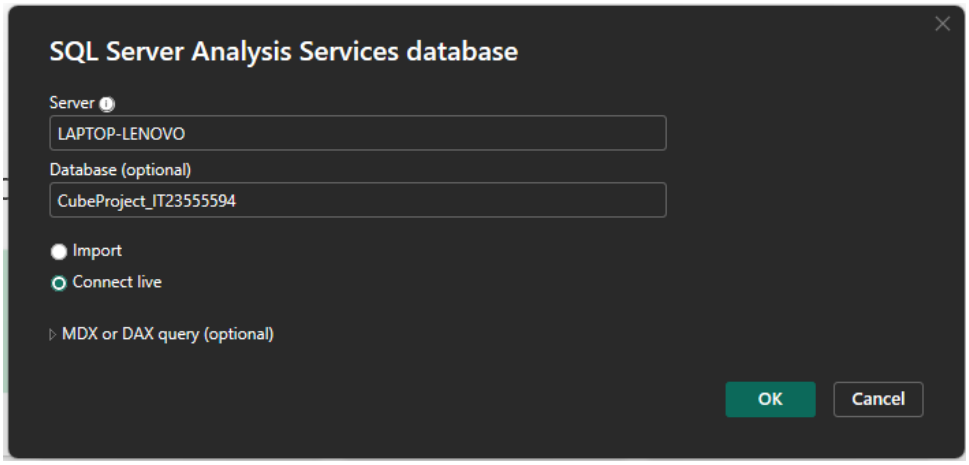
Step 4: Power BI Dashboard Development

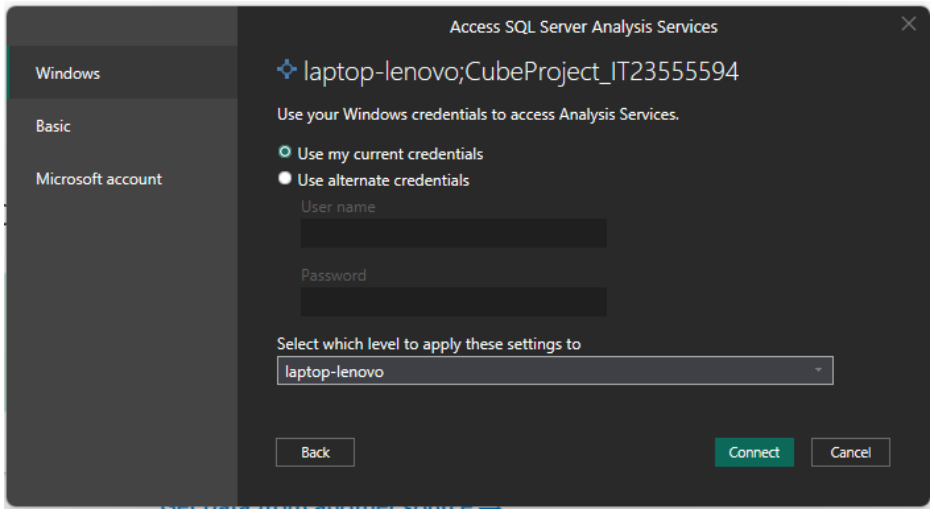
Description:

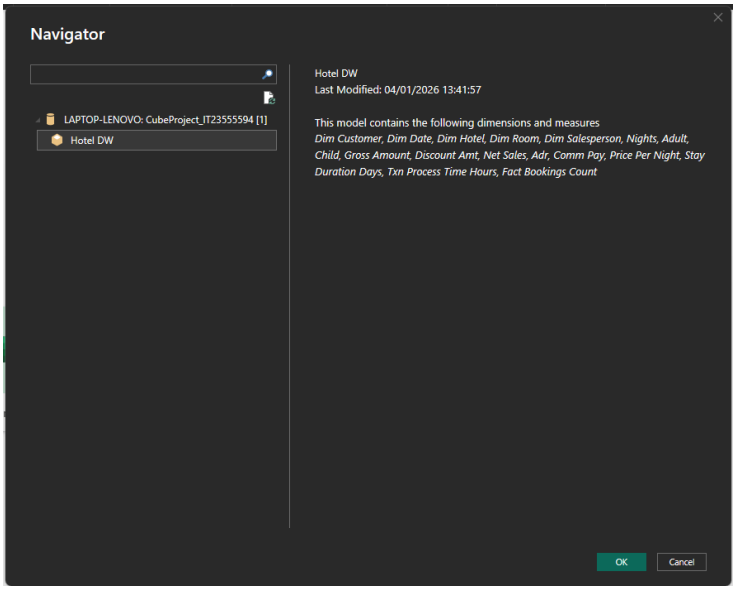
The final phase of the project involved developing an interactive analytical dashboard using **Power BI Desktop**. To ensure data integrity and leverage the multi-dimensional logic built in the previous steps, a **Live Connection** was established to the SSAS Cube hosted on LAPTOP-LENOVO. This ensures that all measures (Net Sales, ADR) and hierarchies (Time) are calculated by the OLAP engine.

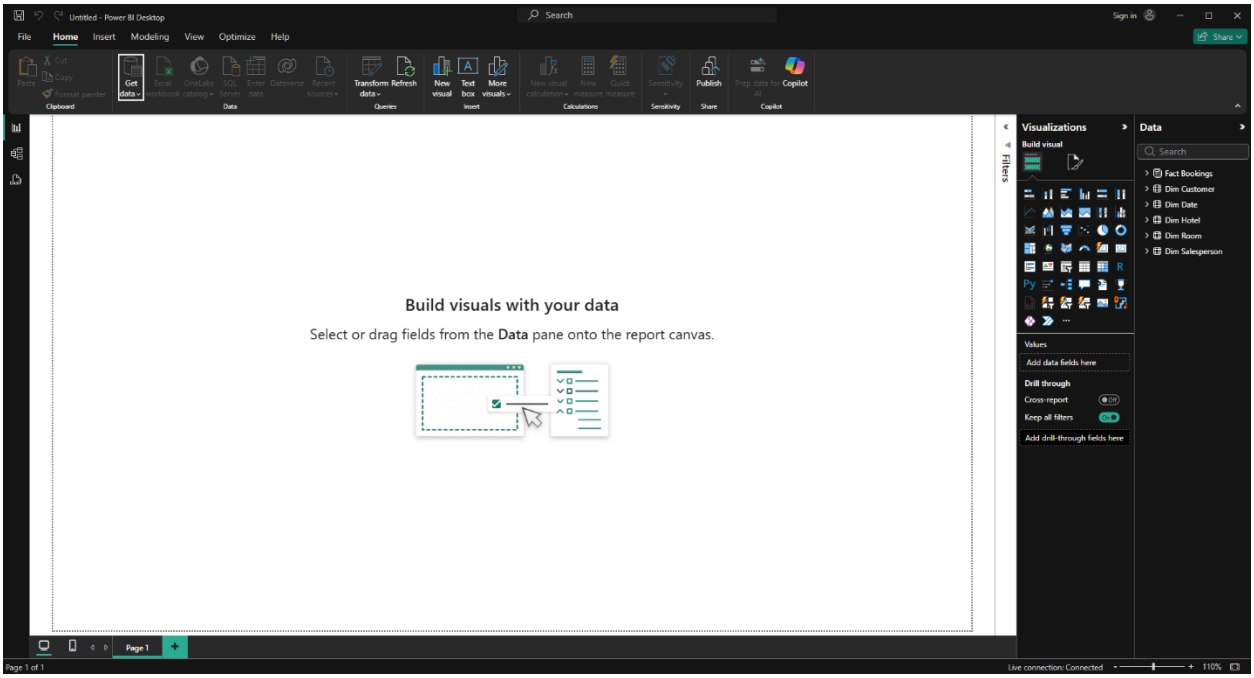
4.1. Connecting to the SSAS Cube

- **Action:** Selected **Get Data > Analysis Services**.
- **Server:** LAPTOP-LENOVO.
- **Connection Mode: Connect Live**.
- **Result:** This allowed the Power BI report to act as a dynamic visualisation layer for the SSAS Cube.



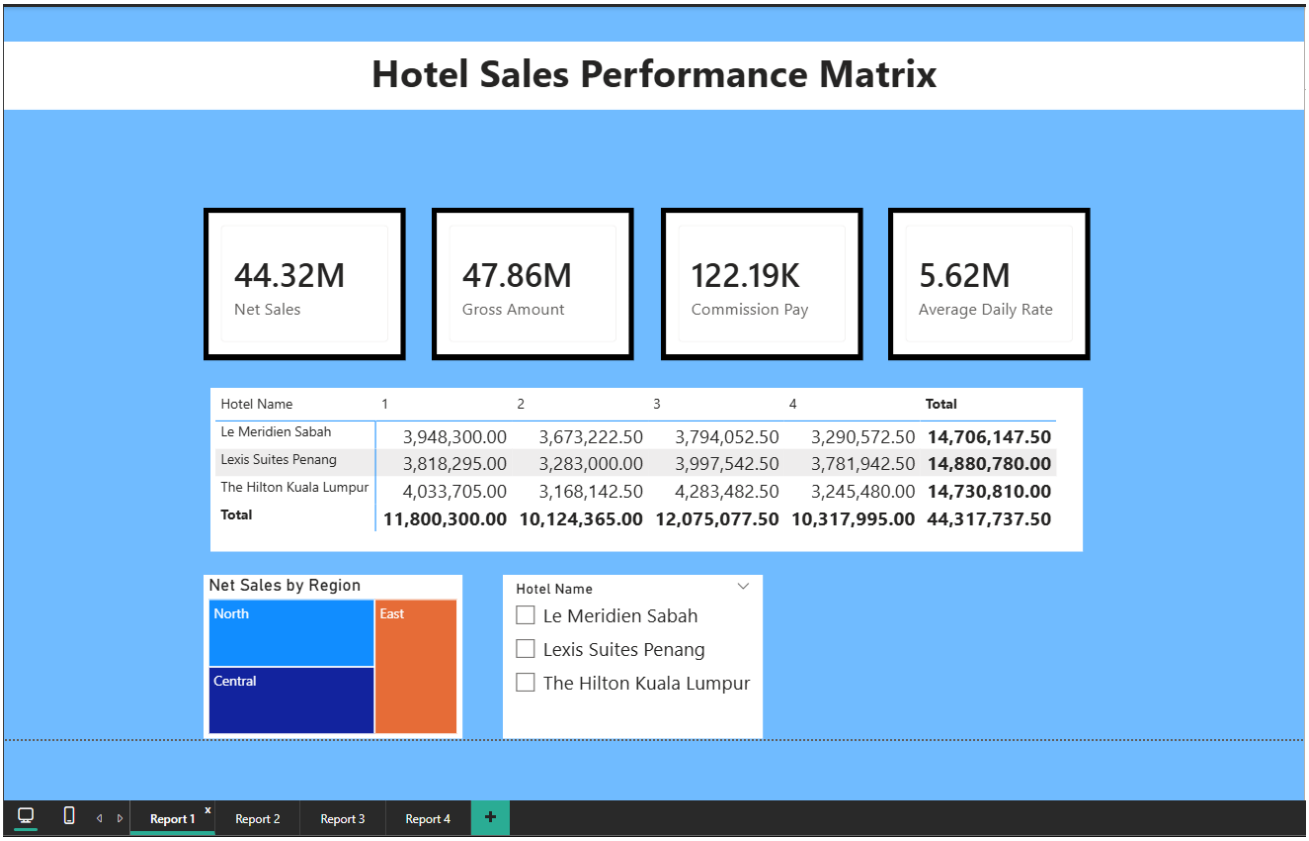



Navigator
LAPTOP-LENOVO: CubeProject_IT23555594 [1]
Hotel DW
Hotel DW
Last Modified: 04/01/2026 13:41:57
This model contains the following dimensions and measures
Dim Customer, Dim Date, Dim Hotel, Dim Room, Dim Salesperson, Nights, Adult, Child, Gross Amount, Discount Amt, Net Sales, Adr, Comm Pay, Price Per Night, Stay Duration Days, Txn Process Time Hours, Fact Bookings Count
OK
Cancel


Untitled - Power BI Desktop
File
Home
Insert
Modeling
View
Optimize
Help
Build visuals with your data
Select or drag fields from the Data pane onto the report canvas.
Visualizations
Data
Fact Bookings
Dim Customer
Dim Date
Dim Hotel
Dim Room
Dim Salesperson
Page 1

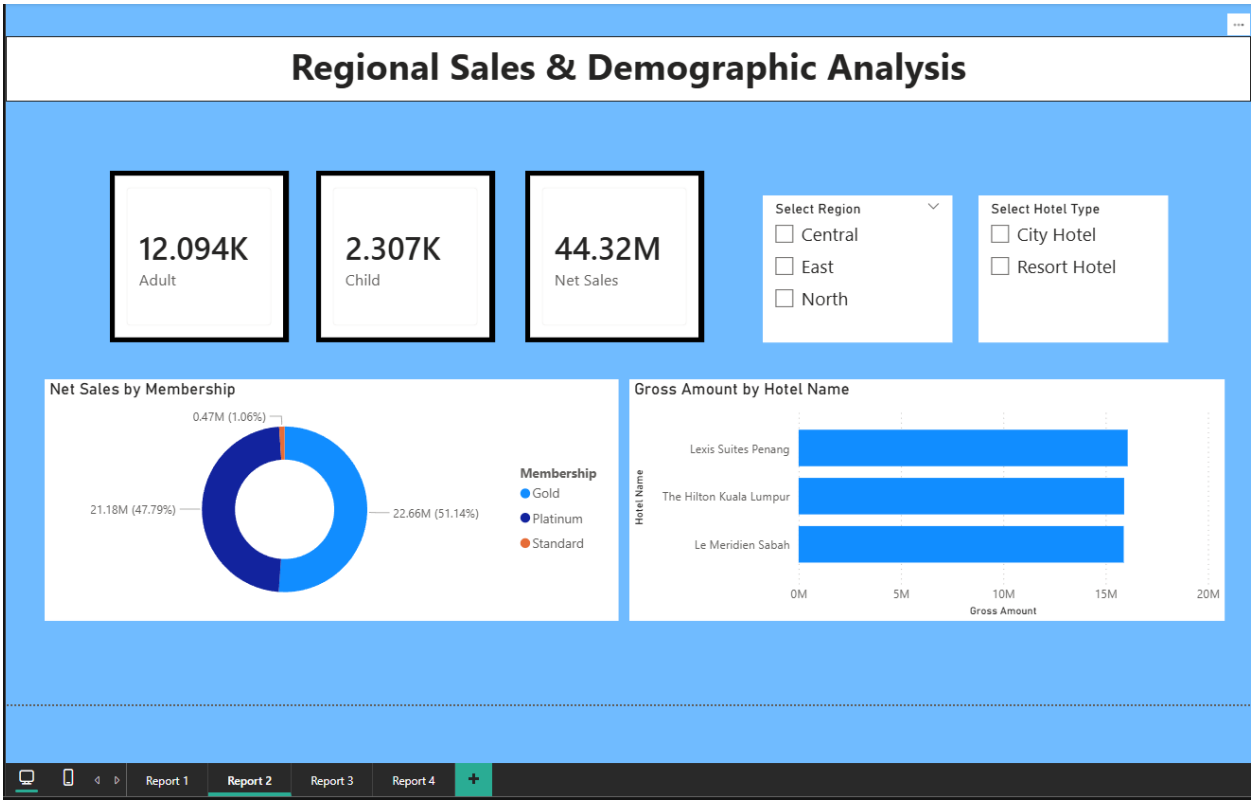
4.2. Report 01: Hotel Sales Performance Matrix

- **Visual Used:** Matrix.
- **Fields:** Hotel Name (Rows), Quarter & Month (Columns), Net Sales (Values).
- **Technical Detail:** This report utilises the **Year-Quarter-Month** hierarchy. The Matrix allows users to expand quarters to see monthly performance totals per hotel.
- **Insight:** Provides a high-level financial summary of all branches for the 2024 fiscal year.



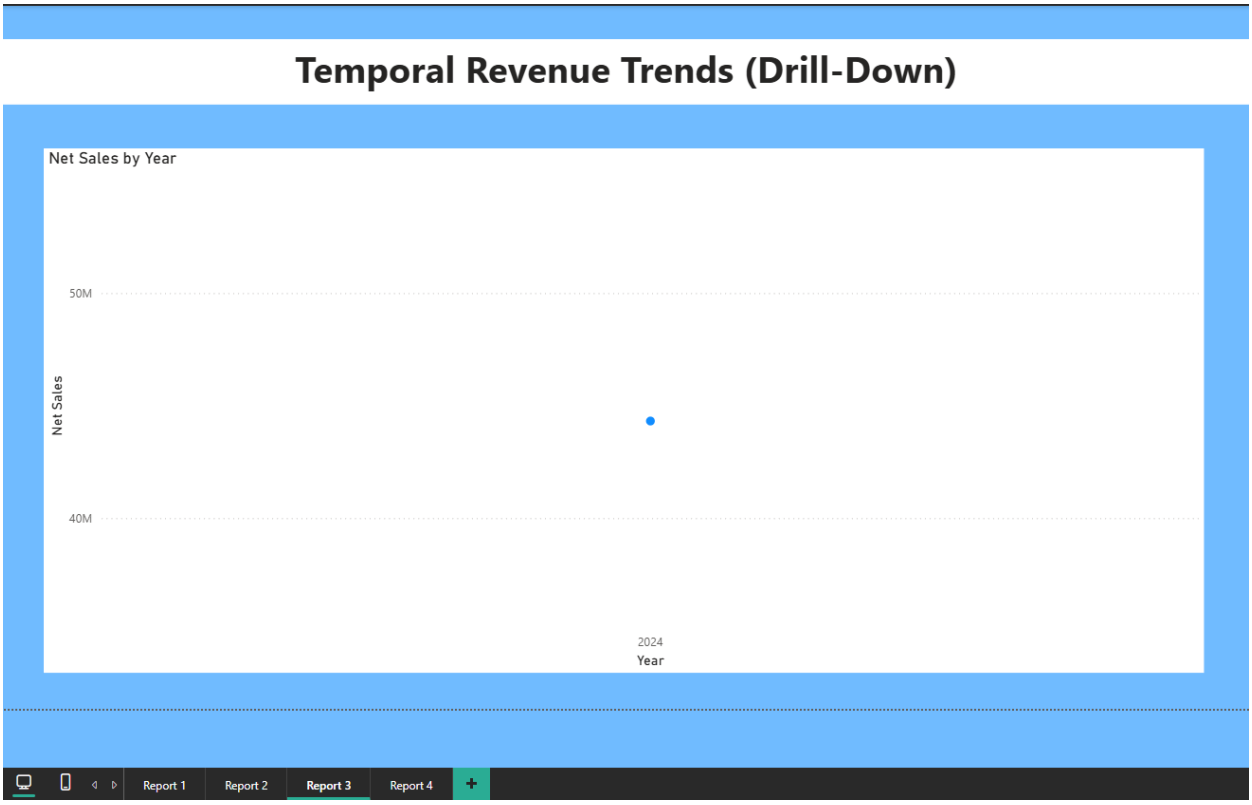
4.3. Report 02: Regional Analysis & Cascading Slicers

- **Visuals Used:** Doughnut Chart, Clustered Bar Chart, and Slicers.
- **Technical Detail:** Implemented **Cascading Slicers**. When a user selects a Region (e.g., Western), the Hotel Type slicer dynamically filters to show only relevant categories.
- **KPIs Included:** Total Net Sales, Adult Count, and Child Count cards.



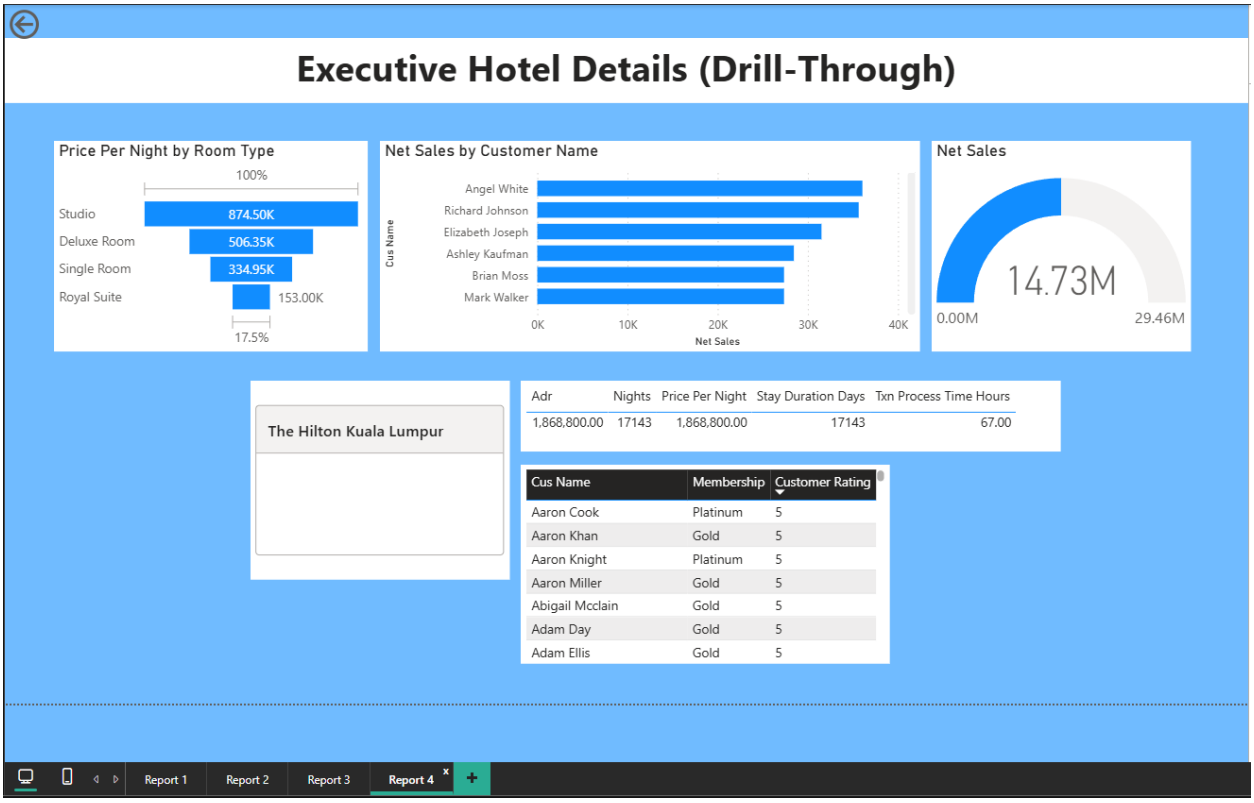
4.4. Report 03: Temporal Revenue Trends (Drill-Down)

- **Visual Used:** Area Chart.
- **Technical Detail:** This report demonstrates the **Drill-Down** capability. By clicking the "Turn on Drill Down" icon, users can navigate from a high-level **Quarterly view** into a detailed **Monthly view** of revenue trends.
- **Insight:** Helps stakeholders identify seasonal peaks and troughs in booking revenue.



4.5. Report 04: Executive Hotel Insights (Drill-Through)

- **Visuals Used:** Gauge Chart, Table, and Multi-row Card (or Table alternative).
- **Technical Detail:** This is a hidden detail page that uses a **Drill-Through filter** on Hotel Name.
- **Action:** Users can right-click a hotel name in Report 01 and select "Drill-through" to land on this page, which automatically filters all visuals for that specific hotel.
- **KPI:** Includes a **Gauge Chart** comparing Net Sales against the 2024 performance target.

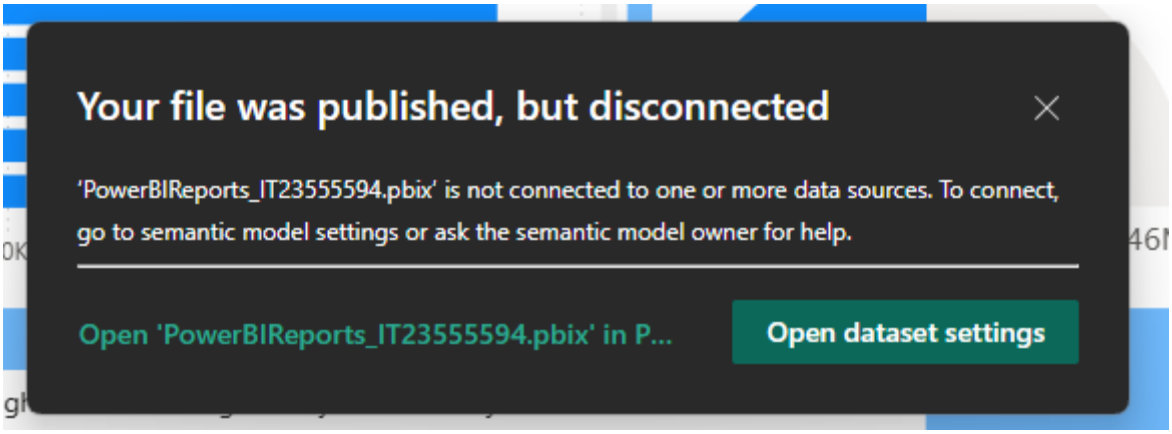


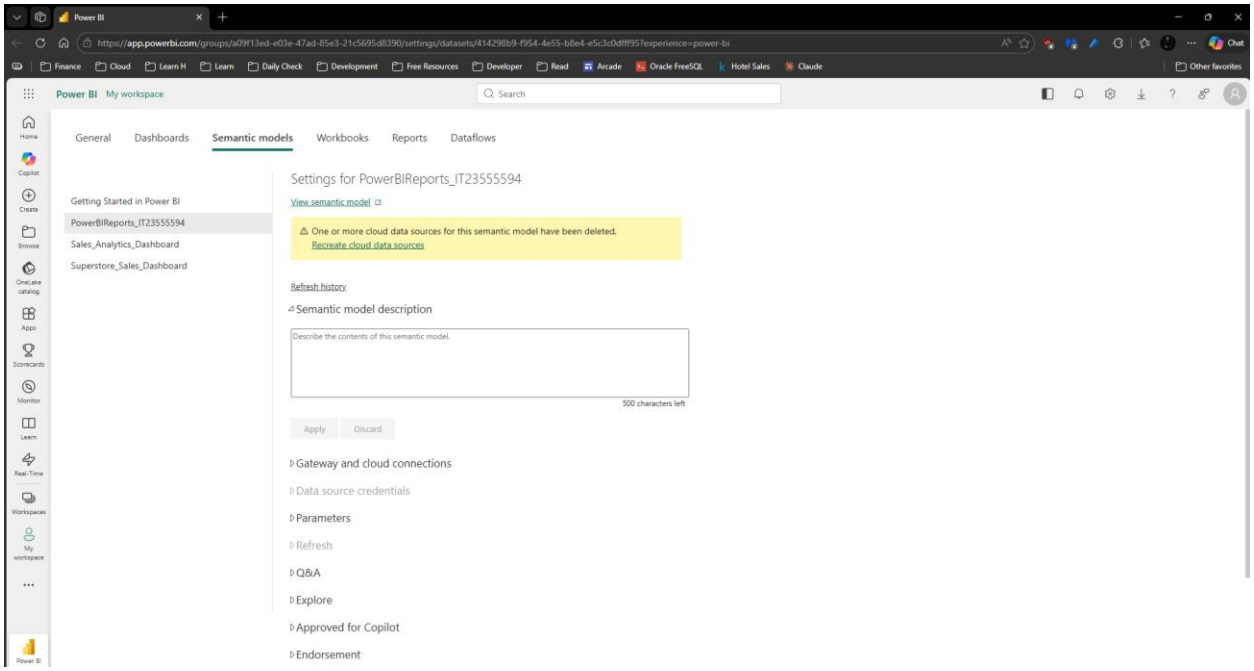
Final Section: Technical Challenges

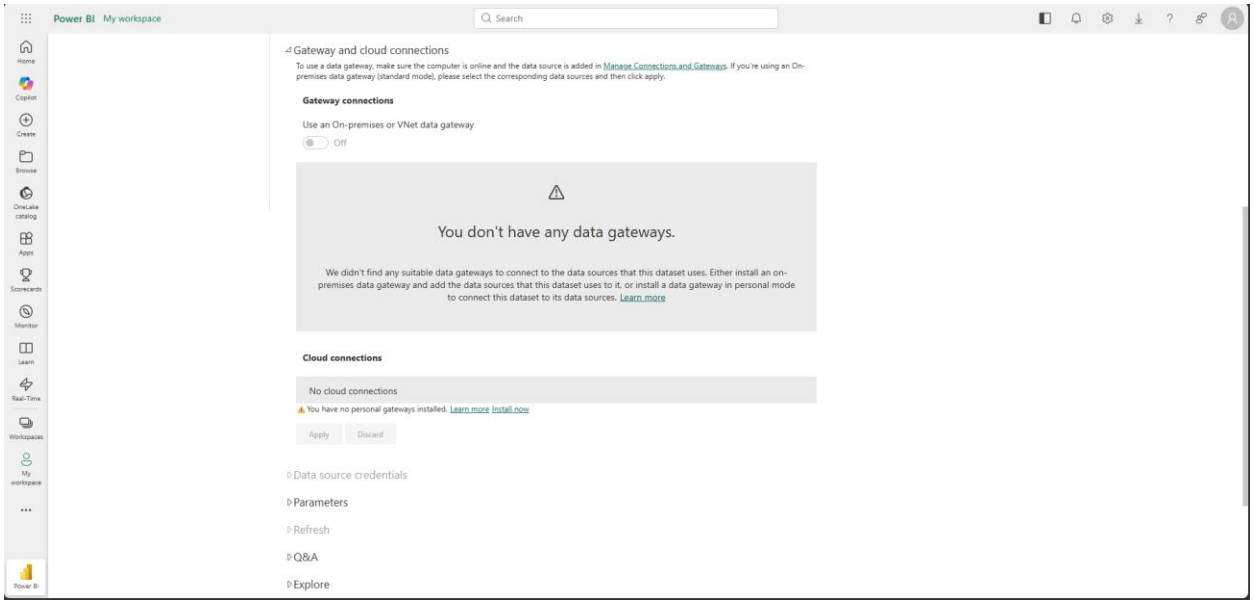
Description:

Upon publishing the report to the **Power BI Service**, a "Disconnected" error was encountered. This is a documented architectural constraint: the Cloud Service requires a **Microsoft On-Premises Data Gateway** to communicate with a local SSAS instance on LAPTOP-LENOVO.

For this assignment, the functionality of the **Live Connection**, including all cube-based measures and hierarchies, has been successfully demonstrated and validated within the Power BI Desktop environment.






Power BI
My workspace
Search
Home
Copilot
Create
Browse
OneLake catalog
Apps
Scorecards
Monitor
Learn
Real-Time
Workspaces
My workspace
Power BI
Gateway and cloud connections
To use a data gateway, make sure the computer is online and the data source is added in Manage Connections and Gateways. If you're using an On-premises data gateway (standard mode), please select the corresponding data sources and then click apply.
Gateway connections
Use an On-premises or VNet data gateway
Off
You don't have any data gateways.
We didn't find any suitable data gateways to connect to the data sources that this dataset uses. Either install an on-premises data gateway and add the data sources that this dataset uses to it, or install a data gateway in personal mode to connect this dataset to its data sources. Learn more
Cloud connections
No cloud connections
You have no personal gateways installed. Learn more Install now
Apply
Discard
Data source credentials
Parameters
Refresh
Q&A
Explore